
On JEPA Isotropy

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We study the empirical risk minimization (ERM) of linear Joint Embedding
2 Predictive Architecture (JEPA) and establish its key conditions for optimality.
3 By casting the embedding bottleneck as a rank constraint, we formulate JEPA as
4 a reduced-rank regression problem. This induces a risk decomposition into an
5 irreducible error and an excess-risk bound. Within the irreducible term, we find
6 the rank of the optimal predictor grows with target heterogeneity and saturates at
7 the bottleneck. Within the excess-risk bound, we find spectral conditioning of the
8 whitened end-to-end predictor governs minimization. Together, these conditions
9 prescribe an isotropic regularizer. Under this regularization, we show such JEPA
10 isotropy subsumes canonical JEPA isotropic embedding designs and hence justifies
11 their empirical successes via ERM. Numerical experiments corroborate our theory.

12 1 Introduction

13 We study the empirical risk minimization (ERM) of Joint-Embedding Predictive Architecture
14 (JEPA) [1, 2] and establish its conditions for optimality. JEPA jointly trains a context encoder,
15 a target encoder, and a predictor so that the predicted target embedding matches the target’s. It has
16 become a prominent self-supervised learning paradigm [2–5]. A key recipe in JEPA is to enforce
17 isotropy on the encoder’s marginal embedding, either through architectures [2–4] or through distribu-
18 tional regularizations [6, 5]. Prior work justifies this recipe by appealing to typical self-supervised
19 methods [7–9], by showing it prevents representational [10, 11] and early-training [12] collapse, or
20 by scoping to specific objectives [5, 13]. More fundamentally, they regularize only the encoder and
21 ignore the predictor, the defining component of JEPA beyond plain Joint-Embedding Architectures
22 (JEAs). As a result, it remains unclear what JEPA’s loss is optimizing. This work targets these gaps
23 through a finite-sample analysis. Specifically, we answer:

24 (Q1) What condition on the encoder and predictor *jointly* minimizes JEPA’s loss?

25 (Q2) How does the rank of the optimal predictor depend on the number and structure of targets?

26 (Q3) Why does encoder isotropy work, and what unifies its existing forms?

27 Casting the embedding bottleneck as a rank constraint, we formulate JEPA as reduced-rank regres-
28 sion [14, 15] and decompose its finite-sample risk into an irreducible population term and an excess
29 term (Theorem 3.15). We answer (Q1) by showing that the excess term is governed by a single
30 condition on the end-to-end prediction. The excess term is minimized at a flat spectrum condition
31 termed *predictive isotropy* (Lemma 4.1 and Theorem 4.2). We answer (Q2) by characterizing the
32 irreducible term through the rank of the optimal predictor. It grows with the heterogeneity of targets
33 and saturates at the embedding bottleneck (Propositions 3.4, 3.8 and 3.9).

34 We thus answer (Q3) from a gauge-invariant risk-minimization view. The predictive isotropy condition
35 for (Q1) acts on the end-to-end prediction. Its SVD supplies a factorization into encoder and predictor,
36 and gauge transformations reparameterize them within the same composition. The risk is gauge-
37 invariant, so any factorization that realizes predictive isotropy minimizes it. Whitening, VICReg,

38 Gaussian-embedding, and LeJEPa [16, 6, 13, 5] each select a gauge toward a specific marginal
 39 embedding (Theorems 4.5 and 4.6). Predictive isotropy subsumes and justifies encoder isotropy.

40 Building on this unification, we derive a predictive-isotropy regularizer that acts only on the end-
 41 to-end prediction from a maximum-entropy principle. Empirically, it improves both test risk and
 42 train-test gap over encoder isotropy on MNIST image prediction. As the gauge-invariant view for
 43 (Q3) predicts, it yields an isotropic-Gaussian embedding as a byproduct. Synthetic experiments
 44 further validate the intermediate claims of our analysis.

45 **Contribution.** Our contributions include:

- 46 • **First ERM Theory of JEPa.** We provide, to our knowledge, the first finite-sample theory of the
 47 JEPa objective. Under linearization, JEPa reduces to reduced-rank regression, yielding a risk
 48 decomposition governed by the spectrum of the whitened end-to-end predictor T_K (Theorem 3.15).
- 49 • **Identifying the Risk Minimizer.** We show that the excess-risk bound is uniquely minimized
 50 when T_K has a flat nonzero singular spectrum. This identifies predictive isotropy as an optimality
 51 condition for JEPa beyond the scope of existing encoder-only designs (Theorem 4.2).
- 52 • **Justification and Unification of Encoder Isotropy.** Our framework gives a unified risk-based
 53 explanation for empirically successful encoder-isotropy recipes, including Whitening, VICReg,
 54 Gaussian-embedding, and LeJEPa. Under predictive isotropy, these methods arise as gauge choices
 55 in the encoder-predictor factorization of T_K (Theorems 4.5 and 4.6).
- 56 • **Regularization Toward the Risk Optimum.** We translate the unique minimizer into a practical
 57 regularizer on the end-to-end prediction (16). Empirically, it reduces test risk and train-test gap,
 58 while inducing isotropic-Gaussian embedding as a natural byproduct rather than a constraint.

59 **Roadmap.** Section 2 introduces preliminaries. Section 3 develops the reduced-rank regression
 60 formulation and risk decomposition. Section 4 establishes predictive isotropy as the unique optimality
 61 condition and shows it subsumes existing encoder-isotropy designs. Section 5 reports numerical
 62 validations. Section 6 concludes. Section B, Section C, and Section D contain extended discussion
 63 and related work, full proofs, and experimental details, respectively.

64 2 Preliminaries

65 In this section, we introduce the notation and background needed for our JEPa analysis.

66 Let $z \in \mathbb{R}^D$ denote a data sample drawn from a distribution $\mathcal{P}$ on $\mathbb{R}^D$. A *mask* is an index set
 67 $m \subseteq [D]$ selecting a subset of coordinates. A *mask distribution* $\mathcal{D}$ is a probability distribution over
 68 tuples $(m_c, m_{t,1}, \dots, m_{t,K})$ specifying one *context block* and $K \geq 1$ *target blocks*.¹ For a sampled
 69 tuple, write $z_{\text{ctx}} := z_{m_c} \in \mathbb{R}^{D_x}$ and $z_{\text{tgt}}^{(k)} := z_{m_{t,k}} \in \mathbb{R}^{D_y^{(k)}}$ for the corresponding context and target
 70 subvectors. JEPa jointly trains an encoder and a predictor to predict the embedding of each target
 71 block from the embedding of the context block.

72 **Architecture.** JEPa consists of three components:

- 73 (i) A *context encoder* $f_\theta : \mathbb{R}^{D_x} \rightarrow \mathbb{R}^d$ mapping the context block to an embedding $x = f_\theta(z_{\text{ctx}})$;
- 74 (ii) A *target encoder (teacher)* $f_{\bar{\theta}} : \mathbb{R}^{D_y^{(k)}} \rightarrow \mathbb{R}^d$ mapping each target to an embedding $y^{(k)} = f_{\bar{\theta}}(z_{\text{tgt}}^{(k)})$;
- 75 (iii) A *predictor* $g_W : \mathbb{R}^d \times \mathcal{Z} \rightarrow \mathbb{R}^d$ that predicts the k -th target embedding $y^{(k)}$ from the context
 76 embedding x and target-side metadata $q_k \in \mathcal{Z}$, i.e., $\hat{y}^{(k)} = g_W(x, q_k)$.

77 Throughout this work, we assume a frozen teacher and a linear end-to-end map.

78 **Assumption 2.1** (Frozen Teacher). The target encoder $f_{\bar{\theta}}$ is fixed during training of the context
 79 encoder f_θ and predictor g_W .

80 **Assumption 2.2** (End-to-end linearity). The end-to-end map $g_W \circ f_\theta : \mathbb{R}^{D_x} \rightarrow \mathbb{R}^{Kd}$ is linear.

81 For explicit matrix-level analysis (Section 3.1-3.5), we consider a specialized linearity assumption.

82 **Assumption 2.3** (Linear factorization). The context encoder is linear, and for each target block
 83 $k \in [K]$ with fixed metadata q_k , the predictor is linear in the context embedding:

$$f_\theta(z_{\text{ctx}}) = Az_{\text{ctx}}, \quad g_W(x, q_k) = W_k x, \quad A \in \mathbb{R}^{d \times D_x}, \quad W_k \in \mathbb{R}^{d \times d}.$$

84 Note Assumption 2.3 implies Assumption 2.2 and serves as the working factorization. Section 4
 85 relaxes it to nonlinear factorizations within the same end-to-end linear map (Lemma 4.4).

¹We call a mask a *block mask* when it selects a contiguous index set. The target blocks need not be disjoint.

86 Under Assumption 2.3, stacking the target embeddings,

$$Y := [(y^{(1)})^\top \ \dots \ (y^{(K)})^\top]^\top \in \mathbb{R}^{Kd}, \quad (1)$$

$$\hat{Y} = \begin{bmatrix} \hat{y}^{(1)} \\ \vdots \\ \hat{y}^{(K)} \end{bmatrix} = Wx, \quad W := \begin{bmatrix} W_1 \\ \vdots \\ W_K \end{bmatrix} \in \mathbb{R}^{Kd \times d}. \quad (2)$$

87 Thus the end-to-end map from context to predicted target embeddings is $WA \in \mathbb{R}^{Kd \times D_x}$.

88 Below we define the JEPA training objective. Unless otherwise stated, all expectations are taken over
89 $z \sim \mathcal{P}$ and $(m_e, m_{t,1}, \dots, m_{t,K}) \sim \mathcal{D}$.

90 **Definition 2.4** (JEPA's loss). Let $x := Az_{\text{ctx}} \in \mathbb{R}^d$ and stack the target embeddings $y^{(k)} := f_{\hat{\theta}}(z_{\text{tgt}}^{(k)})$
91 into Y as in (1), and the predictor blocks W_k into W as in (2). The K -target JEPA's loss is

$$\mathcal{L}_K(W, A) := \frac{1}{K} \mathbb{E}[\|Y - Wx\|^2]. \quad (3)$$

92 The composition WA acts on the raw context x only through its row space, a low-dimensional
93 subspace of $\mathbb{R}^{D_x}$ that captures the directions JEPA effectively operates on. We name it next.

94 **Definition 2.5** (Predictive subspace). A linear subspace $V \subseteq \mathbb{R}^{D_x}$ is called a *predictive subspace* if
95 $V = \text{row}(WA)$ for some admissible predictor WA under Assumption 2.3. We write $\mathcal{V}_r$ for the set
96 of predictive subspaces of dimension r .

97 Denote the population second moments $\Sigma_{XX} := \mathbb{E}[xx^\top] \in \mathbb{R}^{d \times d}$, $\Sigma_{Y^{(k)}X} := \mathbb{E}[y^{(k)}x^\top] \in \mathbb{R}^{d \times d}$
98 for $k \in [K]$, and their stack $\Sigma_{YX} := \mathbb{E}[Yx^\top] = [\Sigma_{Y^{(1)}X}^\top, \dots, \Sigma_{Y^{(K)}X}^\top]^\top \in \mathbb{R}^{Kd \times d}$. Assume
99 $\Sigma_{XX} \succ 0$ (non-degenerate context embedding).

100 3 Main Theory

101 The section analyses JEPA's risk in two stages: what it is, and what minimizes it. Sections 3.1 to 3.3
102 show how target structure and the embedding bottleneck saturate the population risk; Section 3.4
103 identifies a condition as the residual lever beyond saturation; Section 3.5 unifies both into a risk
104 decomposition. Sections 4 and 4.2 identify its unique minimizer and translate it into a regularizer.

105 3.1 JEPA as Reduced-Rank Regression

106 We show that, under the frozen-teacher and linearity (Assumptions 2.1 and 2.3), JEPA reduces to a
107 reduced-rank regression problem. For $r \leq d$, consider the rank-constrained JEPA problem

$$W_r^* \in \arg \min_{W \in \mathbb{R}^{Kd \times d}} \mathcal{L}_K(W, A) \quad \text{subject to} \quad \text{rank}(W) \leq r, \quad (4)$$

108 where $\mathcal{L}_K(W, A)$ is the JEPA's loss in Definition 2.4.

109 **Proposition 3.1** (JEPA as reduced-rank regression). *Under Assumptions 2.1 and 2.3, the optimization*
110 *problem (4) is exactly the reduced-rank regression problem in Definition A.1 with $p = d$, $q = Kd$,*
111 *predictor $x \in \mathbb{R}^d$, and response $Y \in \mathbb{R}^{Kd}$.*

112 Let $B_{\text{OLS}} := \Sigma_{YX} \Sigma_{XX}^{-1}$ from (17). By Lemma A.2, the closed-form solution of (4) is $W_r^* =$
113 $B_{\text{OLS}} \Sigma_{XX}^{1/2} V_r V_r^\top \Sigma_{XX}^{-1/2}$, where $V_r \in \mathbb{R}^{d \times r}$ collect the top- r eigenvalues of $\Sigma_{XX}^{1/2} B_{\text{OLS}}^\top B_{\text{OLS}} \Sigma_{XX}^{1/2}$.

114 Below we study when increasing the number of target blocks can enlarge the predictive structure
115 accessible from the context embedding x .

116 3.2 Multi-Target Masking and Rank Growth

117 The JEPA predictor in (4) is subject to two rank bottlenecks: the explicit constraint r , and an intrinsic
118 bottleneck induced by the data distribution through x . Lemma 3.2 makes the latter precise:

119 **Lemma 3.2** (Rank of the JEPA solution). *Let W_r^* be the solution to Proposition 3.1. Then*

$$\text{rank}(W_r^*) = \min\{r, \text{rank}(\Sigma_{YX})\}. \quad (5)$$

120 *Proof.* See Section C.1 for a detailed proof. $\square$

121 Lemma 3.2 identifies $\text{rank}(\Sigma_{YX})$ as the intrinsic bottleneck predictive rank: adding target blocks
122 enlarges the predictive structure only insofar as it raises this rank. To study how $\text{rank}(\Sigma_{YX})$ depends

on the choice and number of target blocks, we aggregate across blocks. Recall $\Sigma_{Y^{(k)}X}$, Σ_{YX} defined in Section 2. It is natural to define the predictor-side Gram matrix

$$G_K := \Sigma_{YX}^\top \Sigma_{YX} = \sum_{k=1}^K \Sigma_{Y^{(k)}X}^\top \Sigma_{Y^{(k)}X} \in \mathbb{R}^{d \times d}. \quad (6)$$

Each summand $\Sigma_{Y^{(k)}X}^\top \Sigma_{Y^{(k)}X}$ is positive semidefinite, so adding a block can only enlarge the range of G_K (equivalently, shrink its kernel). Throughout, we write $\lambda_i(\cdot)$ and $\sigma_i(\cdot)$ for the i -th largest eigenvalue and singular value, and $\|\cdot\|_{\text{op}}$ for the operator norm. By (6), $\lambda_i(G_K) = \sigma_i(\Sigma_{YX})^2$ for every i , and each summand $\Sigma_{Y^{(k)}X}^\top \Sigma_{Y^{(k)}X} \succeq 0$.

We introduce a predictive rank (Definition 3.3, Proposition 3.4, and Corollary 3.5) measuring *how large* the predictable subspace is, and a heterogeneity level (Definition 3.6 and Proposition 3.8) measuring *how well* it can be recovered from samples.

Definition 3.3 (Predictive rank). For a given K , the *predictive rank* is $r_*(K) := \text{rank}(\Sigma_{YX})$.

Since $\Sigma_{XX} \succ 0$, $G_K = \Sigma_{YX}^\top \Sigma_{YX} \in \mathbb{R}^{d \times d}$, one also has $r_*(K) = \text{rank}(G_K) \leq d$. Intuitively, $r_*(K)$ is the predictive subspace dimension (Definition 2.5). We next study how it depends on K .

Proposition 3.4 (Rank growth). Let $G_K := \Sigma_{YX}^\top \Sigma_{YX}$, $r_*(K) := \text{rank}(\Sigma_{YX})$ for $K \geq 1$. Then:

1. **Monotonicity.** $r_*(K+1) \geq r_*(K)$.
2. **Strict Growth under Complementarity.** $r_*(K+1) > r_*(K)$ if $\ker(G_K) \not\subseteq \ker(\Sigma_{Y^{(K+1)}X})$.

Proof. See Section C.2 for a detailed proof. $\square$

Corollary 3.5 (Saturation). There exists K^* such that $r_*(K) = r_*(K^*) \leq d$ for every $K \geq K^*$.

Besides the predictive rank, it is useful to track how far G_K is from being rank-deficient on its range.

Definition 3.6 (Heterogeneity level). The heterogeneity level λ_{het} is the smallest nonzero eigenvalue of G_K , equivalently $\lambda_{r_*(K)}(G_K)$.

By the identity $G_K = \Sigma_{YX}^\top \Sigma_{YX}$, this is the squared smallest nonzero singular value of Σ_{YX}

$$\sigma_{r_*(K)}(\Sigma_{YX}) = \sqrt{\lambda_{\text{het}}}. \quad (7)$$

Remark 3.7 (Heterogeneity as complementarity). Rank increases only if the new block cross-covariance $\Sigma_{Y^{(K+1)}X}$ is nonzero on $\ker(G_K)$. Equivalently, the new block must reveal directions of X invisible to the previously accumulated structure; highly overlapping or redundant target blocks fail this condition. In this sense, heterogeneity means complementarity among blockwise cross-covariances, not merely a larger number of targets.

The next proposition shows that λ_{het} improves how well the predictive subspace of Σ_{YX} is identified.

Proposition 3.8 (Heterogeneity controls predictive-subspace identifiability). Assume the family $\{\Sigma_{Y^{(k)}X}\}_{k=1}^K$ has heterogeneity level $\lambda_{\text{het}} > 0$ (Definition 3.6). Let $\hat{\Sigma}_{YX}$ be an empirical cross-covariance estimator and let $V_{r_*(K)}$, $\hat{V}_{r_*(K)}$ denote the top- $r_*(K)$ right singular subspaces of Σ_{YX} and $\hat{\Sigma}_{YX}$ respectively. Suppose $E_{YX} := \{\|\hat{\Sigma}_{YX} - \Sigma_{YX}\|_{\text{op}} \leq \eta_n^{YX}(\delta)\}$ has probability at least $1 - \delta$ for some $\eta_n^{YX}(\delta) < \sqrt{\lambda_{\text{het}}}$. Then, on E_{YX} ,

$$\|\sin \Theta(V_{r_*(K)}, \hat{V}_{r_*(K)})\|_{\text{op}} \leq \frac{\eta_n^{YX}(\delta)}{\sqrt{\lambda_{\text{het}}} - \eta_n^{YX}(\delta)}.$$

Proof. See Section C.3 for a detailed proof. $\square$

Under the **half-gap condition** $\eta_n^{YX}(\delta) \leq \sqrt{\lambda_{\text{het}}}/2$, the bound simplifies to $2\eta_n^{YX}(\delta)/\sqrt{\lambda_{\text{het}}}$. Thus stronger heterogeneity tightens the $\sin \Theta$ bound at a fixed cross-covariance perturbation level. A standard sub-Gaussian instantiation for $\eta_n^{YX}(\delta)$ follows from Lemma A.4.

3.3 Capacity Limits of the Embedding Bottleneck

This subsection studies the architectural bottleneck itself. Since JEPA predicts through a d -dimensional representation $x = Az_{\text{ctx}}$, its end-to-end predictor from the raw context block to the stacked target embedding must factor through a rank at most d map. This induces an approximation limit even when the raw context contains richer predictive structure. We denote

$$Z := z_{\text{ctx}} \in \mathbb{R}^{D^x}, \quad \Sigma_{ZZ} := \mathbb{E}[ZZ^\top], \quad \Sigma_{YZ} := \mathbb{E}[YZ^\top],$$

164 and assume $\Sigma_{ZZ} \succ 0$. Define the unconstrained population linear predictor from raw context
 165 to stacked targets by $B_{\text{OLS}}^{\text{raw}} := \Sigma_{YZ} \Sigma_{ZZ}^{-1} \in \mathbb{R}^{Kd \times D_x}$. The next result formalizes the irreducible
 166 approximation error imposed by the d -dimensional embedding bottleneck.

167 **Proposition 3.9** (Capacity lower bound from the embedding bottleneck). *Let $T_K := B_{\text{OLS}}^{\text{raw}} \Sigma_{ZZ}^{1/2}$ and*
 168 *$\sigma_1(T_K) \geq \sigma_2(T_K) \geq \dots \geq 0$ denote its singular values. Then for every $(W, A) \in \mathbb{R}^{Kd \times d} \times \mathbb{R}^{d \times D_x}$,*

$$\mathbb{E} [\|Y - WAZ\|_2^2] \geq \mathbb{E} [\|Y - B_{\text{OLS}}^{\text{raw}} Z\|_2^2] + \sum_{j>d} \sigma_j(T_K)^2. \quad (8)$$

169 *Proof.* See Section C.4 for a detailed proof. $\square$

170 The tail $\sum_{j>d} \sigma_j(T_K)^2$ is architecturally irreducible. No choice of (W, A) can recover predictive
 171 content beyond the top- d singular directions of T_K . This sharpens the saturation of Section 3.2, since
 172 the embedding dimension d is a ceiling that extra target blocks cannot bypass.

173 3.4 Post-Saturation Gains via Eigenvalue Balancing

174 By Corollary 3.5, $r_*(K)$ saturates at some $r_*(K) \leq d$, so extra targets cannot enlarge the predictive
 175 subspace. However, its effectiveness still depends on spectral conditioning. Here we study how this
 176 condition affects prediction. Given $T_K = B_{\text{OLS}}^{\text{raw}} \Sigma_{ZZ}^{1/2}$ (Proposition 3.9) and $H_K := T_K^\top T_K$, define

$$\kappa_K^2 := \frac{\lambda_d(H_K)}{\lambda_1(H_K)} = \left(\frac{\sigma_d(T_K)}{\sigma_1(T_K)} \right)^2 \in (0, 1]. \quad (9)$$

177 We focus on the saturated rank- d regime over a finite target window. Here $r_*(K) = d$ means that the
 178 predictive rank has reached the architectural ceiling, and $\text{rank}(T_K) = d$ makes the rank- d Wedin
 179 gap equal to $\sigma_d(T_K)$, so the relative gap is $\kappa_K = \sigma_d(T_K)/\sigma_1(T_K)$. To separate conditioning from
 180 estimator noise, define the relative perturbation event

$$E_T(K; \rho_n) := \left\{ \|\hat{T}_K - T_K\|_{\text{op}} \leq \rho_n(\delta) \|T_K\|_{\text{op}} \right\}. \quad (10)$$

181 For any $r \leq d$, let $V_{K,r}, \hat{V}_{K,r} \in \mathcal{V}_r$ denote the top- r right singular subspaces of T_K and $\hat{T}_K$,
 182 respectively. To translate conditioning-driven subspace stability into excess risk, we assume a
 183 Lipschitz relation between the two. For a predictive subspace $V \in \mathcal{V}_r$, define

$$\mathcal{R}_K(V) := \frac{1}{K} \inf_{W, A: \text{row}(WA) \subseteq V} \mathbb{E} [\|Y - WAZ\|_2^2] \quad (11)$$

184 as the optimal per-target risk among predictors using V . The factor $1/K$ removes trivial growth in
 185 target dimension, so variation in $\mathcal{R}_K$ reflects predictive quality rather than aggregation.

186 **Assumption 3.10** (Subspace-Lipschitz excess risk). For a fixed target count K and rank $r \leq d$, let
 187 $\mathcal{R}_K^*(r) := \mathcal{R}_K(V_{K,r})$. There exists a constant $L > 0$ such that

$$\mathcal{R}_K(\hat{V}_{K,r}) - \mathcal{R}_K^*(r) \leq L \|\sin \Theta(V_{K,r}, \hat{V}_{K,r})\|_{\text{op}}.$$

188 Based on Assumption 3.10, we obtain a risk decomposition of the per-target risk into a population
 189 floor and a finite-sample excess term, with the derivation deferred to Section A.3.

190 **Proposition 3.11** (Post-saturation risk decomposition). *Fix $K \in \{K^*, \dots, K_{\max}\}$ with $r_*(K) =$*
 191 *$\text{rank}(T_K) = d$ and $\kappa_K^2 \geq \kappa_0^2 + \gamma(K - K^*)$ for some $\kappa_0, \gamma > 0$. Suppose $\mathbb{P}(E_T(K; \rho_n)) \geq 1 - \delta$*
 192 *and the half-gap condition (22) holds. Then, under Assumption 3.10, on $E_T(K; \rho_n)$,*

$$\mathcal{R}_K(\hat{V}_{K,d}) \leq \underbrace{\frac{1}{K} (\mathbb{E} [\|Y - B_{\text{OLS}}^{\text{raw}} Z\|_2^2] + \mathcal{E}_{\text{tail}}(d))}_{\mathcal{R}_K^*(d), \text{ population rank-}d \text{ risk}} + \underbrace{\frac{2L\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K - K^*)}}}_{\text{excess risk}}. \quad (12)$$

193 *Proof.* See Section C.5 for a detailed proof. $\square$

194 The excess-risk term in (12) is governed by two finite-sample effects: shrinkage in n at fixed K
 195 (Remark 3.12), and post-saturation conditioning gains in K at fixed n (Remark 3.13).

196 **Remark 3.12** (Sample complexity). The risk bound is controlled by the relative perturbation level
 197 $\rho_n(\delta)$. If a task-specific estimator analysis gives $\rho_n(\delta) = \tilde{O}(\sqrt{r_{\text{rel}}/n})$ over the finite comparison
 198 window, then $\mathcal{R}_K(\hat{V}_{K,d}) - \mathcal{R}_K^*(d) \leq \varepsilon$ is obtained at $n = \tilde{O}(L^2 r_{\text{rel}} / ((\kappa_0^2 + \gamma(K - K^*))\varepsilon^2))$.

199 **Remark 3.13** (Second saturation). Corollary 3.5 establishes a population saturation of $r_*(K)$. Even
 200 when this saturation occurs at the architectural ceiling $r_* = d$, Proposition 3.11 establishes a second,
 201 finite-sample saturation: recovery continues to improve as new blocks balance the eigenvalue-
 202 conditioning of H_K , until κ_K^2 stops improving, with absolute ceiling 1.

203 3.5 Risk Reduction in Multi-Target JEPA

204 The previous subsections identify two mechanisms: heterogeneity of $\{\Sigma_{Y^{(k)}X}\}_{k=1}^K$ controls the
 205 predictive rank and improves finite-sample identifiability (Propositions 3.4 and 3.8); post-saturation
 206 recovery improves by balancing the spectrum of $T_K := B_{\text{OLS}}^{\text{raw}} \Sigma_{ZZ}^{1/2}$ (Proposition 3.11). The next
 207 theorem unifies them. Let $V_{K,r}, \widehat{V}_{K,r} \in \mathcal{V}_r$ be the top- r right singular subspaces of T_K and $\widehat{T}_K$.

208 For fixed K and $r \leq d$, define the **absolute gap** $\Delta_{K,r}^{\text{abs}} := \sigma_r(T_K) - \sigma_{r+1}(T_K)$, with the convention
 209 $\sigma_{r+1}(T_K) = 0$ when $r = \text{rank}(T_K)$, and the **relative gap** $\bar{\Delta}_{K,r} := \Delta_{K,r}^{\text{abs}} / \|T_K\|_{\text{op}}$. Proposition 3.11
 210 requires a half-gap condition (22); for the general rank- r decomposition, we only require the relative
 211 perturbation level to be smaller than the corresponding relative Wedin gap:

212 **Assumption 3.14** (Gap-compatible relative perturbation). For the fixed target count K and rank
 213 $r \leq d$, the event $E_T(K; \rho_n)$ in (10) satisfies $\mathbb{P}(E_T(K; \rho_n)) \geq 1 - \delta$ for some $\rho_n(\delta) < \bar{\Delta}_{K,r}$.

214 **Theorem 3.15** (Unified risk decomposition). Under Assumptions 3.10 and 3.14, on $E_T(K; \rho_n)$,

$$\mathcal{R}_K(\widehat{V}_{K,r}) \leq \underbrace{\frac{1}{K} \left(\mathbb{E} [\|Y - B_{\text{OLS}}^{\text{raw}} Z\|^2] + \sum_{j>r} \sigma_j(T_K)^2 \right)}_{\mathcal{R}_K^+(r), \text{ population rank-}r \text{ risk}} + \underbrace{L \frac{\rho_n(\delta)}{\bar{\Delta}_{K,r} - \rho_n(\delta)}}_{\text{excess-risk}}. \quad (13)$$

215 *Proof.* See Section C.6 for a detailed proof. $\square$

216 **Remark 3.16** (Risk-reduction levers). The bound (13) isolates two controllable levers given fixed
 217 targets and context. *Capacity*: the population rank- r floor $\frac{1}{K} \sum_{j>r} \sigma_j(T_K)^2$ vanishes once $r \geq$
 218 $r_*(K)$, and is bottlenecked by the architectural ceiling $r = d$; the residual tail $\frac{1}{K} \sum_{j>d} \sigma_j(T_K)^2$
 219 persists whenever $d < r_*(K)$. *Conditioning*: the finite-sample term shrinks as the retained gap
 220 grows **relative to the perturbation envelope**. In the saturated rank- d case, this is monitored by
 221 $\kappa_K^2 = \lambda_d(H_K) / \lambda_1(H_K)$ at a fixed relative perturbation level. The remaining task-predictability floor
 222 $\frac{1}{K} \mathbb{E} \|Y - B_{\text{OLS}}^{\text{raw}} Z\|^2$ is irreducible and tracked by $r_*(K)$ (Definition 3.3).

223 4 Predictive Isotropy

224 Theorem 3.15 decomposes the empirical risk of JEPA. In this section, we bridge its risk minimization
 225 to a predictive isotropy condition (Section 4.1), and then show that this condition is gauge-invariant,
 226 thus subsuming and unifying existing encoder-isotropy designs into a regularization (Section 4.2).

227 4.1 Unique Risk Minimizer

228 Recall the stacked context $Z \in \mathbb{R}^{D_x}$ from Section 3.3, the linear context encoder A from Assump-
 229 tion 2.3, and the whitened end-to-end predictor T_K with $H_K := T_K^\top T_K$ from Proposition 3.9.

230 The object of interest is the end-to-end prediction $P_K := T_K \tilde{Z}$, i.e., $P_K = B_{\text{OLS}}^{\text{raw}} Z$ written in
 231 whitened coordinates. The rank- d JEPA end-to-end prediction WAZ approximates P_K under the
 232 embedding bottleneck (8). Whereas LeJEPA's SIGReg regularizes the law of the embedding AZ
 233 toward an isotropic Gaussian prior [5], we target the prediction-aware analogue on P_K .

234 **Lemma 4.1** (Predictive isotropy). Let $H_K = T_K^\top T_K$. Assuming $\Sigma_{ZZ} \succ 0$, $\mathbb{E}[\tilde{Z}] = 0$, and
 235 $\text{rank}(T_K) = d$, the end-to-end prediction $P_K = T_K \tilde{Z}$ has isotropic covariance on its rank- d
 236 predictive range if and only if

$$\lambda_1(H_K) = \lambda_2(H_K) = \cdots = \lambda_d(H_K). \quad (14)$$

237 *Proof.* See Section C.7 for a detailed proof. $\square$

238 We now show predictive isotropy (Lemma 4.1) is the unique minimizer of the excess-risk bound.

239 **Theorem 4.2** (Predictive isotropy minimizes excess-risk bound). Fix $\tau > 0$, a **relative perturbation**
 240 **level** $\rho_n(\delta)$, and consider rank- d operators T_K satisfying $\text{tr}(H_K) = \|T_K\|_F^2 = \tau$ and $\rho_n(\delta) < \kappa_K$.
 241 Among all such operators, the isotropic spectrum $\lambda_1(H_K) = \cdots = \lambda_d(H_K) = \tau/d$ uniquely
 242 minimizes the rank- d **Wedin excess-risk bound** $\rho_n(\delta) / (\kappa_K - \rho_n(\delta))$ appearing in Theorem 3.15.

243 *Proof.* See Section C.8 for a detailed proof. $\square$

244 The recovery-bound optimum in Theorem 4.2 is an instance of a broader principle, see Section A.4.

245 4.2 Regularization

246 Lemma 4.1 admits a distributional reading that places the present subsection one step further down
 247 the JEPA pipeline than LeJEPA’s SIGReg. Recall that for a measure μ and a map f , the pushforward
 248 $f_{\#}\mu$ denotes the law of $f(X)$ when $X \sim \mu$. SIGReg regularizes the embedding pushforward
 249 $\mathcal{L}(AZ) = A_{\#}\mathcal{L}(Z)$ toward an isotropic Gaussian prior [5]; the prediction-aware analogue is the
 250 end-to-end predictive pushforward

$$\mathcal{L}(P_K) = (T_K)_{\#}\mathcal{L}(\tilde{Z}). \quad (15)$$

251 *Remark 4.3* (Relation to encoder isotropic regularization). **Covariance isotropy of the predictive**
 252 **pushforward $\mathcal{L}(P_K)$ (15) on its rank- d range is exactly the eigenvalue-balancing condition of $H_K =$**
 253 **$T_K^{\top}T_K$ (Lemma 4.1). Full Gaussianity is a stronger maximum-entropy completion (Lemma A.9).**

254 Within Lemma 4.1’s condition, the encoder-predictor factorization remains a structural choice. We
 255 specify one that bridges the predictive isotropy with the embedding isotropy of canonical JEPA
 256 paradigms [5]. Recall the SVD of the whitened end-to-end predictor, $T_K = U_K S_K V_K^{\top}$, and let
 257 $V_K^{\top}\tilde{Z} \in \mathbb{R}^d$ be the SVD coordinate of the whitened context.

258 **Lemma 4.4** (Gauge invariance of the rank- d optimum). *Let a gauge $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be any invertible*
 259 *map. Define a encoder-predictor factorization (Λ_G, F_G)*

$$\Lambda_G := G \circ V_K^{\top} \Sigma_{ZZ}^{-1/2}, \quad F_G := U_K S_K \circ G^{-1}.$$

260 *Then $F_G \circ \Lambda_G$ realizes the population rank- d end-to-end prediction P_K for every choice of G .*

261 By Lemma 4.4, predictive isotropy $H_K = \alpha I_d$ is gauge-invariant, so it constrains the end-to-end
 262 map T_K without committing to any embedding distribution. The gauge G is therefore free to be
 263 chosen for downstream purposes. In particular, it can be selected to realize either of the two canonical
 264 encoder regularization paradigms in JEPA: covariance isotropy (whitening [16] or VICReg-style [6])
 265 or Gaussian isotropy (LeJEPA [5], SIGReg-style).

266 **Theorem 4.5** (Subsumption of covariance isotropy). *For any orthogonal gauge $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the*
 267 *encoder-predictor factorization (Λ_G, F_G) in Lemma 4.4 induces an isotropic covariance embedding*
 268 *$\text{Cov}(\Lambda_G(Z)) = I_d$.*

269 *Proof.* See Section C.9 for a detailed proof. $\square$

270 Theorem 4.5 realizes covariance isotropy with a linear gauge. We next show that a nonlinear gauge
 271 can realize the strictly stronger property of isotropic-Gaussian embedding under a mild assumption.

272 **Theorem 4.6** (Subsumption of Gaussian isotropy). *Assume $V_K^{\top}\tilde{Z}$ has absolutely continuous law on*
 273 *$\mathbb{R}^d$. Then there exists an invertible gauge $G^* : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that the encoder-predictor factorization*
 274 *(Λ_{G^*}, F_{G^*}) in Lemma 4.4 induces an isotropic Gaussian embedding $\Lambda_{G^*}(Z) \sim \mathcal{N}(0, I_d)$.*

275 *Proof Sketch.* Existence of G^* follows from Brenier’s theorem [17], since the law of $V_K^{\top}\tilde{Z}$ admits
 276 a density on $\mathbb{R}^d$ by hypothesis and $\mathcal{N}(0, I_d)$ admits a density on $\mathbb{R}^d$. The encoder-predictor pair
 277 (Λ_{G^*}, F_{G^*}) then satisfies the conclusion by construction (Lemma 4.4). $\square$

278 When the context is Gaussian, the gauge in Theorem 4.6 collapses to the identity.

279 **Corollary 4.7** (Linear-Gaussian special case). *If the context follow a Gaussian distribution $Z \sim$*
 280 *$\mathcal{N}(0, \Sigma_{ZZ})$, the encoder-predictor factorization reduces to the linear pair*

$$\Lambda_I = V_K^{\top} \Sigma_{ZZ}^{-1/2}, \quad F_I = U_K S_K.$$

281 *Proof.* See Section C.10 for a detailed proof. $\square$

282 Subsumption in Theorems 4.5 and 4.6 means the converse fails: encoder isotropy alone does not
 283 imply predictive isotropy, since it is compatible with arbitrary predictor conditioning $\kappa_K^2 \in (0, 1]$.
 284 Moreover, by Lemma 4.4, every choice of G realizes the same end-to-end prediction $P_K = T_K \tilde{Z}$
 285 with the same optimality condition in Theorem 4.2, so selecting a gauge in Theorems 4.5 and 4.6
 286 costs nothing against the excess-risk bound.

287 We now turn this principle into a regularizer. The excess-risk bound in Theorem 3.15 motivates
 288 isotropy of the gauge-free predictive object H_K . The subsumption results in Theorems 4.5 and 4.6
 289 further show that, under such regularization, encoder isotropy need not be imposed separately, nor
 290 must a gauge in Lemma 4.4 be chosen manually.

291 **Maximum-entropy regularization induces isotropic-Gaussian embedding.** The excess-risk bound
 292 prescribes only the spectral condition (14) on H_K , equivalently covariance isotropy of P_K on its
 293 rank- d predictive range. This places candidate regularizers on a two-axis frontier: *fidelity*, whether
 294 $\mathcal{R} = 0$ implies (14), and *commitment*, how much distributional structure $\mathcal{R} = 0$ enforces beyond it.
 295 The least-committing faithful choice is the maximum-entropy law satisfying predictive isotropy. At
 296 trace τ , Lemma A.9 shows that this law is uniquely $\mathcal{N}(0, \frac{\tau}{d}I_d)$ on the intrinsic predictive range.

297 Therefore, we propose a maximum-entropy regularizer for the intrinsic predicted output $\hat{P}_K^{\text{int}} :=$
 298 $\hat{U}_K^\top \hat{W} \hat{A} Z \in \mathbb{R}^d$, where $\hat{U}_K \in \mathbb{R}^{Kd \times d}$ spans the learned rank- d predictive range. By the Cramér-
 299 Wold theorem [18], in its hyperspherical form [5, Lemma 3], the target law $\hat{P}_K^{\text{int}} \sim \mathcal{N}(0, I_d)$ is
 300 equivalent to $u^\top \hat{P}_K^{\text{int}} \sim \mathcal{N}(0, 1)$ for every $u \in \mathbb{S}^{d-1}$. We approximate this all-direction condition by
 301 sampling $|\mathbb{A}|$ random unit vectors $\mathbb{A} \subset \mathbb{S}^{d-1}$, projecting the predictor output onto each direction, and
 302 averaging the standard-normal Gaussianity statistics T (Epps-Pulley [19]):

$$\mathcal{R}_{\text{PI}}(\hat{P}_K^{\text{int}}; \mathbb{A}) := \frac{1}{|\mathbb{A}|} \sum_{u \in \mathbb{A}} T(\{u^\top \hat{p}_n\}_{n=1}^N), \quad (16)$$

303 where $\hat{p}_n := \hat{U}_K^\top \hat{W} \hat{A} z_n$ is the intrinsic predicted output on the n -th sample.

304 By Lemma 4.1, the covariance part of $\mathcal{N}(0, I_d)$ enforces predictive isotropy on the nonzero spectrum
 305 of H_K , which minimizes the excess-risk bound (Theorem 4.2). The Gaussian part is its maximum-
 306 entropy completion. (16) is therefore the predictive analogue of LeJEPa’s SIGReg [5], constraining
 307 $\mathcal{L}(\hat{P}_K^{\text{int}})$ rather than $\mathcal{L}(AZ)$. Practically, (16) acts on the gauge-free predictive object H_K , so it
 308 requires no manual gauge choice in Lemma 4.4. Together with Theorems 4.5 and 4.6, it justifies and
 309 unifies the two encoder regularization paradigms in canonical JEPA [16, 6, 5] as factorized instances
 310 of the same predictive-isotropy principle.

311 5 Experimental Studies

312 5.1 Synthetic Experiment

313 **Setup.** We first use frozen-teacher linear experiments with known population operators to isolate the
 314 mechanisms in the theory from nonlinear optimization effects. Full setup details are in Section D.1.

315 **Models and baselines.** The learned model is the finite-sample RRR solution. Rank d instantiates the
 316 JEPA bottleneck. Population operators and unconstrained OLS serve as baselines.

317 **Metrics.** Risk panels use the per-target normalization in (11). Diagnostics match the quantities in
 318 Theorem 3.15 and the rank, heterogeneity, and conditioning results. Definitions are in Section D.1.

319 **Results.** The appendix figures Figure 3, Figures 4 to 6 validate the **reduction**, pre-saturation, and
 320 bottleneck claims. **The factorized linear predictor realizes the RRR solution**; complementary target
 321 stacks increase predictive rank, larger λ_{het} improves finite-sample identifiability, and the rank-
 322 d bottleneck induces the spectral-tail floor predicted by Propositions 3.4, 3.8 and 3.9. **Figure 7**
 323 **independently checks Theorem 3.15: with $r < r_*(K)$, the population spectral tail is nonzero, the**
 324 **retained-gap condition is non-vacuous, and the calibrated excess term bounds the observed finite-**
 325 **sample risk.** The main synthetic diagnostic is Figure 1. It tests Proposition 3.11 in the regime
 326 $r_*(K) = d$, where additional target blocks cannot create new rank. The experiment allocates new
 327 blocks to weak predictive directions. As $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ increases, the observed excess risk
 328 and per-target risk decrease. Figure 8 tests Lemma 4.1 and Theorem 4.2: at fixed rank and trace,
 329 flatter spectra of $H_K = T_K^\top T_K$ improve recovery and reduce the balanced-reference-stack excess
 330 risk. The gauge-factorization diagnostic in Figure 10 checks that the encoder-isotropy claims in
 331 Section 4.2 are factorization effects rather than new end-to-end prediction constraints.

332 5.2 Realistic Experiment

333 **Setup.** We test the practical regularizer (16) in learned nonlinear image inpainting tasks. In both tasks
 334 the context is the left half of a digit image and the target is the right half. Figure 2 shows the task.

335 **Data.** The small-data diagnostic uses the `sklearn` 8×8 digit dataset. The scaled diagnostic uses
 336 standard 28×28 MNIST. All pixels are normalized to $[0, 1]$.

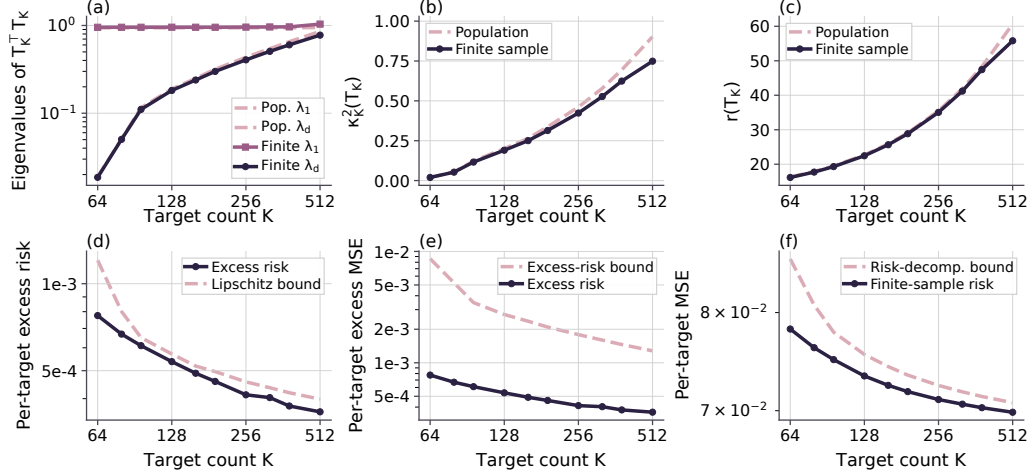


Figure 1: **Post-saturation conditioning.** Given $r_*(K) = d$, allocating targets to weak directions increases κ_K^2 and reduces excess risk and per-target risk. Panel (d) plots the calibrated Lipschitz bound $\tilde{L} \|\sin \Theta_T\|_{\text{op}}$.

Table 1: **Real-image regularizer diagnostics.** MSE and gap columns are reported in 10^{-2} units, and RMSE is reported as a percentage of the normalized $[0, 1]$ pixel range. Gap is test MSE minus train MSE. Foreground MSE is reported only for MNIST, where background pixels dominate the right-half target.

Task	Objective	Test MSE	RMSE (%)	Test/Train Gap	FG MSE	Time
8×8 digits	No reg.	9.23±0.17	30.4±0.3	8.54±0.17	-	9.03±0.20 s (1.00×)
8×8 digits	Enc. reg.	7.86±0.13	28.0±0.2	6.29±0.14	-	12.55±0.11 s (1.39×)
8×8 digits	Pred. reg.	6.00±0.03	24.5±0.1	2.00±0.07	-	12.23±0.04 s (1.35×)
8×8 digits	Pred.+Enc.	6.09±0.04	24.7±0.1	2.27±0.08	-	15.80±0.13 s (1.75×)
MNIST CNN	No reg.	3.68±0.02	19.19±0.04	2.81±0.05	12.90±0.07	69.52±4.23 s (1.00×)
MNIST CNN	Enc. reg.	3.65±0.02	19.11±0.05	2.68±0.05	12.75±0.07	55.39±0.13 s (0.80×)
MNIST CNN	Pred. reg.	3.43±0.02	18.53±0.04	2.13±0.06	11.92±0.06	53.98±0.07 s (0.78×)
MNIST CNN	Pred.+Enc.	3.47±0.01	18.63±0.04	2.12±0.05	12.06±0.05	39.08±4.30 s (0.56×)

337 **Models.** The small-data task uses an MLP encoder-predictor, and MNIST uses a convolutional
338 encoder-decoder. The prediction-side regularizer is applied to the predictor representation as in (16);
339 architecture and optimizer details are in Section D.2.

340 **Baselines.** We compare four objectives: no regularization, Encoder Gaussianity, prediction-side
341 Gaussianity, and their combination. Encoder Gaussianity is the standard representation-side control.
342 Prediction-side Gaussianity is our theory-motivated regularizer in (16), and the combined objective is
343 an ablation testing whether Encoder Gaussianity improves on prediction-side Gaussianity.

344 **Metrics.** We report target-half pixel MSE and RMSE. For MNIST we also report foreground MSE,
345 computed on non-background right-half pixels as most right-half pixels are background. Gaussianity
346 and effective-rank diagnostics are reported in Figures 11 and 12; prediction risk is the primary
347 outcome.

348 **Results.** Figure 2 visualizes the digit-half prediction task, and Table 1 reports the regularizer diagnos-
349 tics. Additional completion examples are shown in Figures 13 and 14. On the low-label 8×8 digit task,
350 prediction-side regularization reduces test MSE from 9.23×10^{-2} to 6.00×10^{-2} and train–test gap
351 from 8.54×10^{-2} to 2.00×10^{-2} . On the scaled MNIST task, prediction-side regularization improves
352 test MSE from 3.68×10^{-2} to 3.43×10^{-2} and foreground MSE from 1.29×10^{-1} to 1.19×10^{-1} . The
353 combined objective adds little, consistent with the subsumption view in Theorems 4.5 and 4.6.

354 6 Concluding Remarks

355 We give a finite-sample ERM analysis of JEPA and identified optimality condition. Casting JEPA
356 as reduced-rank regression (Proposition 3.1), we decompose its risk into population and excess
357 terms (Theorem 3.15). The excess term is uniquely minimized by a flat-spectrum condition termed
358 *predictive isotropy* (Theorem 4.2). The population term is controlled by the optimal predictor’s rank,
359 which grows with target heterogeneity and saturates at the embedding bottleneck (Propositions 3.4,
360 3.8 and 3.9). Because the risk is gauge-invariant, predictive isotropy pins down the end-to-end map
361 but leaves its encoder–predictor factorization free; existing encoder-isotropy recipes each fix a gauge
362 realizing it (Lemma 4.4 and Theorems 4.5 and 4.6). Acting on the end-to-end map, we derive a

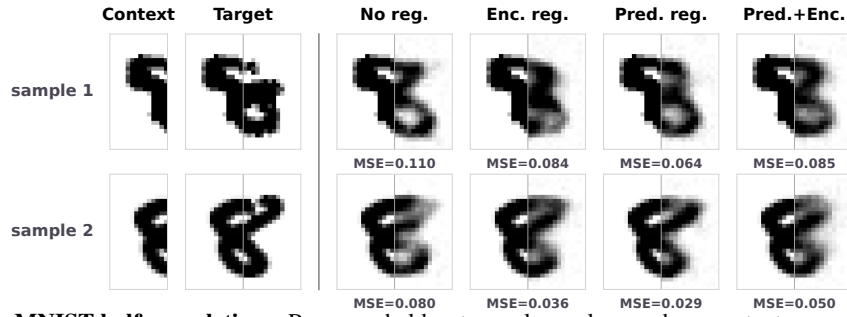


Figure 2: **MNIST half-completions.** Rows are held-out samples; columns show context, ground truth, and candidate completions. Cell labels report per-sample target-half pixel MSE.

363 predictive-isotropy regularizer (16) that subsumes isotropic-Gaussian embedding without fixing a
 364 gauge. We defer discussion with related work to Section B.

References

- [1] Yann LeCun et al. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1):1–62, 2022.
- [2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15619–15629, 2023.
- [3] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. V-jepa: Latent video prediction for visual representation learning. 2023.
- [4] Adrien Bardes, Jean Ponce, and Yann LeCun. Mc-jepa: A joint-embedding predictive architecture for self-supervised learning of motion and content features. *arXiv preprint arXiv:2307.12698*, 2023.
- [5] Randall Balestriero and Yann LeCun. Lejepa: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.
- [6] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021.
- [7] Jean-Bastien Grill, Florian Strub, Florian Althé, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, 2020.
- [8] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021.
- [9] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR, 2020.
- [10] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.
- [11] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning*, pages 10268–10278. PMLR, 2021.
- [12] Sofiane Mansouri et al. Understanding early collapse in predictive world-model pretraining. *OpenReview*, 2026.
- [13] Randall Balestriero, Nicolas Ballas, Mike Rabbat, and Yann LeCun. Gaussian embeddings: How jepas secretly learn your data density. *arXiv preprint arXiv:2510.05949*, 2025.
- [14] Theodore Wilbur Anderson. Estimating linear restrictions on regression coefficients for multivariate normal distributions. *The Annals of Mathematical Statistics*, pages 327–351, 1951.
- [15] Alan Julian Izenman. Reduced-rank regression for the multivariate linear model. *Journal of multivariate analysis*, 5(2):248–264, 1975.
- [16] Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self-supervised representation learning. In *ICML*, 2021.
- [17] Cédric Villani et al. *Optimal transport: old and new*, volume 338. Springer, 2009.
- [18] Harald Cramér and Herman Wold. Some theorems on distribution functions. *Journal of the London Mathematical Society*, s1-11(4):290–294, 1936.

- 411 [19] T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.
- 412
- 413 [20] Gregory C Reinsel and Raja P Velu. *Multivariate reduced-rank regression*. Springer, 1998.
- 414 [21] Vladimir Koltchinskii and Karim Lounici. Concentration inequalities and moment bounds for
- 415 sample covariance operators. *Bernoulli*, pages 110–133, 2017.
- 416 [22] Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT*
- 417 *Numerical Mathematics*, 12(1):99–111, 1972.
- 418 [23] A Wayne Roberts and Dale E Varberg. *Convex Functions: Convex Functions*, volume 57.
- 419 Academic Press, 1974.
- 420 [24] Albert W Marshall, Ingram Olkin, and Barry C Arnold. Inequalities: theory of majorization
- 421 and its applications. 1979.
- 422 [25] Nikunj Saunshi, Orestis Plevrakis, Sanjeev Arora, Mikhail Khodak, and Hrishikesh Khande-
- 423 parkar. A theoretical analysis of contrastive unsupervised representation learning. In *International conference on machine learning*, pages 5628–5637. PMLR, 2019.
- 424
- 425 [26] Christopher Tosh, Akshay Krishnamurthy, and Daniel Hsu. Contrastive estimation reveals topic
- 426 posterior information to linear models. *Journal of Machine Learning Research*, 22(281):1–31,
- 427 2021.
- 428 [27] Jeff Z HaoChen, Colin Wei, Adrien Gaidon, and Tengyu Ma. Provable guarantees for self-
- 429 supervised deep learning with spectral contrastive loss. *Advances in neural information processing systems*, 34:5000–5011, 2021.
- 430
- 431 [28] Jason D Lee, Qi Lei, Nikunj Saunshi, and Jiacheng Zhuo. Predicting what you already know
- 432 helps: Provable self-supervised learning. *Advances in Neural Information Processing Systems*,
- 433 34:309–323, 2021.
- 434 [29] Yuandong Tian, Lantao Yu, Xinlei Chen, and Surya Ganguli. Understanding self-supervised
- 435 learning with dual deep networks. *arXiv preprint arXiv:2010.00578*, 2020.
- 436 [30] Bobak T Kiani, Randall Balestriero, Yubei Chen, Seth Lloyd, and Yann LeCun. Joint embedding
- 437 self-supervised learning in the kernel regime. *arXiv preprint arXiv:2209.14884*, 2022.
- 438 [31] Vivien Cabannes, Bobak Kiani, Randall Balestriero, Yann LeCun, and Alberto Bietti. The ssl
- 439 interplay: Augmentations, inductive bias, and generalization. In *International conference on machine learning*, pages 3252–3298. PMLR, 2023.
- 440
- 441 [32] Etai Littwin, Omid Saremi, Madhu Advani, Vimal Thilak, Preetum Nakkiran, Chen Huang,
- 442 and Joshua Susskind. How japa avoids noisy features: The implicit bias of deep linear self
- 443 distillation networks. *Advances in Neural Information Processing Systems*, 37:91300–91336,
- 444 2024.
- 445 [33] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-
- 446 supervised learning via redundancy reduction. In *International conference on machine learning*,
- 447 pages 12310–12320. PMLR, 2021.
- 448 [34] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank.
- 449 *Psychometrika*, 1(3):211–218, 1936.

Appendix

451	A Supplementary Theoretical Backgrounds	14
452	A.1 Reduced-Rank Regression	14
453	A.2 Useful Lemmas	14
454	A.3 Intermediate Derivation of Post-Saturation Recovery	15
455	A.4 Maximum-Entropy Laws	16
456	B Related Works and Discussion	17
457	C Proofs of Main Text	18
458	C.1 Proof of Rank of Reduced-Rank Solution	18
459	C.2 Proof of Rank Growth	18
460	C.3 Proof of Heterogeneity and Identifiability	19
461	C.4 Proof of Capacity Lower Bound	19
462	C.5 Proof of Post-Saturation Risk Decomposition	20
463	C.6 Proof of Unified Risk Decomposition	20
464	C.7 Proof of Predictive Isotropy	20
465	C.8 Proof of Predictive Isotropy Minimize the Excess-Risk Bound	20
466	C.9 Proof of Covariance Isotropy Subsumption	20
467	C.10 Proof of Linear Gaussian Gauge	21
468	D Additional Experimental Details	21
469	D.1 Synthetic Experiment Details	21
470	D.2 Real-Image Regularizer Details	28

Impact Statement

By the theoretical nature of this work, we do not anticipate any negative social impact.

LLM Usage Disclosure

We used large language models (LLMs) to aid and polish writing, such as improving clarity, grammar, and conciseness. We also used LLMs for retrieval and discovery, for example, literature-search assistance to identify potential missing related work. In addition, LLM-based coding assistants were used as basic code assistance to help implement the numerical validation experiments (e.g., plotting and data pipelines). All technical content, proofs, experiments, and results are original contributions by the authors.

480 A Supplementary Theoretical Backgrounds

481 A.1 Reduced-Rank Regression

482 Given paired random vectors $x \in \mathbb{R}^p$ and $y \in \mathbb{R}^q$, the population squared-loss regression problem

$$\min_{B \in \mathbb{R}^{q \times p}} \mathbb{E}[\|y - Bx\|^2] \quad (17)$$

483 admits the closed-form ordinary least squares (OLS) minimizer $B_{\text{OLS}} := \Sigma_{YX} \Sigma_{XX}^{-1}$ when $\Sigma_{XX} \succ 0$.

484 Reduced-rank regression (RRR) [15, 20] restricts the minimization in (17) to $\text{rank}(B) \leq r$.

485 **Definition A.1** (Reduced-rank regression problem). Given paired $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$ as predictor and
486 response random vectors, for a prescribed rank $r \leq \min(p, q)$, the reduced-rank regression is

$$B_{\text{RRR}} \in \arg \min_{B \in \mathbb{R}^{q \times p}} \mathbb{E}[\|y - Bx\|^2] \quad \text{subject to} \quad \text{rank}(B) \leq r. \quad (18)$$

487 The constraint $\text{rank}(B) \leq r$ means the linear predictor factors through an r -dimensional latent
488 subspace. The following lemma gives a closed-form solution.

489 **Lemma A.2** (RRR solution, Theorem 2.2 of [20]). Let $V_r \in \mathbb{R}^{p \times r}$ collect eigenvectors corresponding
490 to the top r eigenvalues of $\Sigma_{XX}^{1/2} B_{\text{OLS}}^\top B_{\text{OLS}} \Sigma_{XX}^{1/2}$. A solution to (18) is

$$B_{\text{RRR}} = B_{\text{OLS}} \Sigma_{XX}^{1/2} V_r V_r^\top \Sigma_{XX}^{-1/2}. \quad (19)$$

491 Equivalently, RRR solution is obtained by taking the best rank- r approximation of $B_{\text{OLS}} \Sigma_{XX}^{1/2}$ and
492 then unwhitening by $\Sigma_{XX}^{-1/2}$. Informally, this means that RRR retains the r predictor-space directions
493 that are most useful for linearly predicting y .

494 *Remark A.3* (Connection to CCA and PCA). Reduced-rank regression is closely related to canonical
495 correlation analysis (CCA): after appropriate whitening, the rank- r solution retains the leading
496 canonical predictive directions linking the predictor space to the response space. As a special case,
497 when $y = x$, the corresponding reduced-rank self-prediction problem recovers the principal subspace
498 of Σ_{XX} , linking RRR to PCA.

499 A.2 Useful Lemmas

500 **Lemma A.4** (Joint-covariance control of cross-covariance noise [21]). Let $(X_i, Y_i)_{i=1}^n$ be i.i.d.
501 samples with $X \in \mathbb{R}^d$ and $Y \in \mathbb{R}^{K^d}$, and assume $U := (Y^\top, X^\top)^\top$ is sub-Gaussian with second
502 moment $M := \mathbb{E}[UU^\top]$. Let $\widehat{M} := n^{-1} \sum_{i=1}^n U_i U_i^\top$ and $\widehat{\Sigma}_{YX} := n^{-1} \sum_{i=1}^n Y_i X_i^\top$. Then $\widehat{\Sigma}_{YX} -$
503 Σ_{YX} is an off-diagonal block of $\widehat{M} - M$. Consequently, for any $\delta \in (0, 1)$, with probability at least
504 $1 - \delta$,

$$\|\widehat{\Sigma}_{YX} - \Sigma_{YX}\|_{\text{op}} \leq C \|M\|_{\text{op}} \left(\sqrt{\frac{r(M) + \log(1/\delta)}{n}} + \frac{r(M) + \log(1/\delta)}{n} \right),$$

505 where $C > 0$ is an absolute constant and $r(M) := \text{tr}(M)/\|M\|_{\text{op}}$ is the effective rank of the joint
506 second-moment matrix.

507 *Proof.* Write

$$U_i := \begin{pmatrix} Y_i \\ X_i \end{pmatrix}, \quad M := \mathbb{E}[UU^\top], \quad \widehat{M} := \frac{1}{n} \sum_{i=1}^n U_i U_i^\top.$$

508 Then

$$M = \begin{pmatrix} \mathbb{E}[YY^\top] & \mathbb{E}[YX^\top] \\ \mathbb{E}[XY^\top] & \mathbb{E}[XX^\top] \end{pmatrix} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix},$$

509 and hence

$$\widehat{M} - M = \begin{pmatrix} \widehat{\Sigma}_{YY} - \Sigma_{YY} & \widehat{\Sigma}_{YX} - \Sigma_{YX} \\ \widehat{\Sigma}_{XY} - \Sigma_{XY} & \widehat{\Sigma}_{XX} - \Sigma_{XX} \end{pmatrix}.$$

510 Let $B := \widehat{\Sigma}_{YX} - \Sigma_{YX}$. For any unit vectors a, b , define $u = (a^\top, 0^\top)^\top$ and $v = (0^\top, b^\top)^\top$. Then
 511 $a^\top B b = u^\top (\widehat{M} - M) v$, so

$$\|B\|_{\text{op}} \leq \|\widehat{M} - M\|_{\text{op}}.$$

512 Applying the effective-rank sample-covariance concentration bound for sub-Gaussian vectors to
 513 $\widehat{M} - M$ gives

$$\|\widehat{M} - M\|_{\text{op}} \leq C\|M\|_{\text{op}} \left(\sqrt{\frac{r(M) + \log(1/\delta)}{n}} + \frac{r(M) + \log(1/\delta)}{n} \right),$$

514 which proves the claim. $\square$

515 **Lemma A.5** (Wedin-type subspace perturbation [22]). *Let $M \in \mathbb{R}^{m \times p}$, let $r \leq \min(m, p)$, and let*
 516 *$\widehat{M}$ be a perturbation with $\|\widehat{M} - M\|_{\text{op}} \leq \epsilon$. Denote by $V_r, \widehat{V}_r$ the top- r right singular subspaces of*
 517 *M and $\widehat{M}$, and define the Wedin gap at level r as $g_r(M) := \sigma_r(M) - \sigma_{r+1}(M)$, with the convention*
 518 *$\sigma_{r+1}(M) := 0$ when $\text{rank}(M) \leq r$. Provided $\epsilon < g_r(M)$,*

$$\|\sin \Theta(V_r, \widehat{V}_r)\|_{\text{op}} \leq \frac{\epsilon}{g_r(M) - \epsilon}.$$

519 *Proof.* Apply Wedin's $\sin \Theta$ theorem [22] to M and $\widehat{M}$ with spectral gap $g_r(M)$. $\square$

520 A.3 Intermediate Derivation of Post-Saturation Recovery

521 This appendix collects the subspace-recovery results that underpin Proposition 3.11 in the body. The
 522 derivation proceeds in two steps. First, a Wedin-type perturbation bound translates the [deterministic](#)
 523 [conditioning hypotheses and the relative perturbation event](#) (10) into a finite-sample $\sin \Theta$ bound on
 524 the top- d right singular subspace of T_K (Proposition A.6). Second, a [half-gap condition](#) exposes the
 525 explicit dependence on the conditioning parameter (Corollary A.7). The body result is then assembled
 526 in Proposition 3.11 by composing this $\sin \Theta$ bound with the [subspace-Lipschitz](#) Assumption 3.10.

527 Step 1: Wedin $\sin \Theta$ bound.

528 **Proposition A.6** (Post-saturation recovery under [relative perturbation](#)). *Fix $K \in \{K^*, \dots, K_{\max}\}$*
 529 *with $r_*(K) = \text{rank}(T_K) = d$ and $\kappa_K^2 \geq \kappa_0^2 + \gamma(K - K^*)$. For any $\delta \in (0, 1)$, suppose*
 530 *$\mathbb{P}(E_T(K; \rho_n)) \geq 1 - \delta$. Then, on $E_T(K; \rho_n)$ (10),*

$$\|\sin \Theta(V_{K,d}, \widehat{V}_{K,d})\|_{\text{op}} \leq \frac{\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K - K^*)} - \rho_n(\delta)}, \quad (20)$$

531 *provided the denominator is positive.*

532 *Proof.* Since $\text{rank}(T_K) = d$, the Wedin gap $g_d(T_K)$ reduces to $\sigma_d(T_K)$. On $E_T(K; \rho_n)$ (10),

$$\|\sin \Theta(V_{K,d}, \widehat{V}_{K,d})\|_{\text{op}} \leq \frac{\rho_n(\delta) \|T_K\|_{\text{op}}}{\sigma_d(T_K) - \rho_n(\delta) \|T_K\|_{\text{op}}}, \quad (21)$$

533 by Lemma A.5. Since $\sigma_d(T_K) = \kappa_K \|T_K\|_{\text{op}}$, division by $\|T_K\|_{\text{op}} > 0$ gives

$$\|\sin \Theta(V_{K,d}, \widehat{V}_{K,d})\|_{\text{op}} \leq \frac{\rho_n(\delta)}{\kappa_K - \rho_n(\delta)}.$$

534 Using $\kappa_K \geq \sqrt{\kappa_0^2 + \gamma(K - K^*)}$ yields (20). $\square$

535 **Step 2: Explicit K -rate via the half-gap condition.** The bound (20) couples the **relative perturbation**
 536 **envelope** and the deterministic floor in a single ratio. To expose the conditioning dependence, we
 537 impose a **half-gap condition** requiring the relative perturbation to be at most half the relative singular-
 538 **value gap**:

$$\rho_n(\delta) \leq \frac{1}{2} \sqrt{\kappa_0^2 + \gamma(K - K^*)}. \quad (22)$$

539 This half-gap condition is verified empirically across the studied K -range in Figure 1.

540 **Corollary A.7** (Finite-window post-saturation rate). *Fix $K \in \{K^*, \dots, K_{\max}\}$ with $r_*(K) =$
 541 $\text{rank}(T_K) = d$ and $\kappa_K^2 \geq \kappa_0^2 + \gamma(K - K^*)$. For any $\delta \in (0, 1)$, suppose $\mathbb{P}(E_T(K; \rho_n)) \geq 1 - \delta$
 542 and the half-gap condition (22) holds. Then, on $E_T(K; \rho_n)$ (10),*

$$\|\sin \Theta(V_{K,d}, \widehat{V}_{K,d})\|_{\text{op}} \leq \frac{2\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K - K^*)}}. \quad (23)$$

543 *Proof.* Substituting the lower bound in (22) into (20) gives (23). $\square$

544 A.4 Maximum-Entropy Laws

545 We recall two standard entropy facts used in the maximum-entropy regularization argument. Let X
 546 be an absolutely continuous random vector in $\mathbb{R}^d$ with density p_X . Its differential entropy is

$$h(X) := - \int_{\mathbb{R}^d} p_X(x) \log p_X(x) dx,$$

547 whenever the integral is well-defined.

548 **Lemma A.8** (Gaussian maximum entropy under fixed covariance). *Let X be an absolutely continuous*
 549 *random vector in $\mathbb{R}^d$ with mean μ and covariance $\Sigma \succ 0$. Then*

$$h(X) \leq \frac{1}{2} \log((2\pi e)^d \det \Sigma),$$

550 *with equality if and only if $X \sim \mathcal{N}(\mu, \Sigma)$.*

551 *Proof.* Let $G \sim \mathcal{N}(\mu, \Sigma)$. Since $D_{\text{KL}}(\mathcal{L}(X) \parallel \mathcal{L}(G)) \geq 0$,

$$-h(X) - \mathbb{E} \log p_G(X) \geq 0.$$

552 Because X and G have the same mean and covariance,

$$-\mathbb{E} \log p_G(X) = \frac{1}{2} \log((2\pi)^d \det \Sigma) + \frac{1}{2} \mathbb{E}[(X - \mu)^\top \Sigma^{-1} (X - \mu)] = \frac{1}{2} \log((2\pi e)^d \det \Sigma).$$

553 Hence the claimed bound follows. Equality holds iff the KL divergence is zero, i.e. iff X has the
 554 Gaussian law. $\square$

555 **Lemma A.9** (Trace-constrained maximum entropy). *Let X be an absolutely continuous random*
 556 *vector in $\mathbb{R}^d$ with mean zero, covariance $C \succ 0$, and $\text{tr}(C) = \tau$. Let $h(X)$ denote its differential*

557 *entropy. Then*

$$h(X) \leq \frac{1}{2} \log((2\pi e)^d \det C) \leq \frac{d}{2} \log\left(2\pi e \frac{\tau}{d}\right).$$

558 *Equality holds if and only if $X \sim \mathcal{N}(0, \frac{\tau}{d} I_d)$.*

559 *Proof.* By Lemma A.8, applied with $\Sigma = C$ and $\mu = 0$,

$$h(X) \leq \frac{1}{2} \log((2\pi e)^d \det C),$$

560 with equality iff $X \sim \mathcal{N}(0, C)$. For the second inequality, arithmetic-geometric mean (AM-GM)
561 inequality gives

$$\det C = \prod_{i=1}^d \lambda_i(C) \leq \left(\frac{1}{d} \sum_{i=1}^d \lambda_i(C) \right)^d = \left(\frac{\tau}{d} \right)^d,$$

562 with equality iff $\lambda_1(C) = \dots = \lambda_d(C) = \tau/d$, i.e., $C = \frac{\tau}{d} I_d$. Combining the equality conditions
563 gives $X \sim \mathcal{N}(0, \frac{\tau}{d} I_d)$. $\square$

564 **Connection to predictive isotropy.** Theorem 4.2 is the recovery-bound instance of the same fixed-
565 trace principle: at fixed signal energy $\text{tr}(H_K) = \tau$, the flat spectrum $(\tau/d, \dots, \tau/d)$ is the unique
566 majorization-minimum among positive rank- d spectra with total mass τ [23, 24]. Equivalently, by
567 AM-GM, it maximizes the log-determinant term $\frac{1}{2} \sum_{i=1}^d \log \lambda_i(H_K)$ among fixed-trace spectra. The
568 finite-sample condition $\kappa_K^2 \rightarrow 1$ is therefore the empirical manifestation of uniform predictive-energy
569 allocation across the rank- d range.

570 B Related Works and Discussion

571 **Finite-sample theory of joint-embedding learning.** Theory for self-supervised representation
572 learning has largely focused on downstream transfer, contrastive objectives, or kernel-regime joint-
573 embedding models. Contrastive analyses give downstream guarantees under latent-class or multi-
574 view assumptions, including labeled-sample-complexity improvements in latent-class settings [25,
575 26]. Spectral analyses further show how augmentation structure can induce useful representation
576 subspaces with linear-probe guarantees [27]. Related pretext-task and non-contrastive analyses
577 explain how paired views, conditional independence, predictors, and training dynamics support
578 downstream representations and collapse avoidance [28, 29]. Kernel-regime analyses characterize
579 how augmentations, inductive bias, and regularization shape SSL generalization and optimal joint-
580 embedding representations [30, 31]. These results establish finite-sample theory for broad SSL
581 settings, but they do not analyze the JEPA predictive objective as an empirical-risk problem with an
582 explicit low-rank predictor bottleneck. Closer to JEPA, recent deep-linear work explains why latent
583 prediction can avoid noisy reconstruction targets by studying training-dynamics implicit bias [32].

584 In contrast, our analysis targets the ERM of JEPA itself. Under linearization, the predictive objective
585 becomes reduced-rank regression (Proposition 3.1), yielding a finite-sample risk decomposition into
586 a population floor and a finite-sample spectral excess term (Theorem 3.15).

587 **Encoder isotropy and anti-collapse.** A central design principle in non-contrastive SSL is to prevent
588 collapse by controlling the encoder embedding distribution. Whitening-based SSL scatters latent
589 features through batch whitening, Barlow Twins drives the cross-correlation matrix toward identity,
590 and VICReg combines invariance with variance and covariance regularization [16, 33, 6]. Recent
591 JEPA-specific work strengthens this encoder-side perspective by promoting isotropic-Gaussian
592 embeddings as the optimal representation distribution [5, 13].

593 Our result changes the primitive object. For JEPA, risk is uniquely controlled by the spectrum of the
594 whitened end-to-end predictor (Lemma 4.1 and Theorem 4.2). Encoder isotropy is therefore a gauge
595 choice in the encoder-predictor factorization of an end-to-end map satisfying predictive isotropy
596 (Lemma 4.4 and Theorems 4.5 and 4.6).

597 **Entropy and Gaussianity.** Our maximum-entropy argument separates the risk condition from its
 598 distributional completion. The finite-sample bound requires spectral balancing of the predictive
 599 covariance (Theorem 4.2), while Gaussianity enters as the least-committing law satisfying this
 600 covariance-level condition. Thus isotropic-Gaussian embeddings arise as the entropy-optimal com-
 601 pletion of predictive isotropy, realizable through a suitable encoder-predictor gauge (Lemma 4.4
 602 and Theorem 4.6). This reframes Gaussian embedding regularization as an encoder-side instance of a
 603 ERM condition, specifically via our regularizer (16). Stronger distributional constraints should be
 604 justified by the loss and task rather than by collapse prevention alone.

605 **Limitations.** Our analysis is deliberately scoped to a linearized JEPA model with a frozen target
 606 encoder. The linear assumption makes the predictor bottleneck explicit and allows the JEPA objective
 607 to reduce to reduced-rank regression, but it does not capture representation-dependent feature learning
 608 in deep encoders. Similarly, the frozen-teacher assumption isolates the ERM geometry of the context
 609 encoder and predictor, but omits the coupled training dynamics induced by EMA or jointly updated
 610 target networks. Therefore, our results should be read as characterizing the local or linearized risk of
 611 JEPA, rather than a complete theory of nonlinear dynamics. Extending predictive isotropy to adaptive
 612 teachers and nonlinear feature learning remains an important direction for future work.

613 C Proofs of Main Text

614 C.1 Proof of Rank of Reduced-Rank Solution

615 *Proof of Lemma 3.2.* Using Lemma A.2 and $B_{\text{OLS}} = \Sigma_{YX} \Sigma_{XX}^{-1}$,

$$W_r^* = \Sigma_{YX} \Sigma_{XX}^{-1/2} V_r V_r^\top \Sigma_{XX}^{-1/2}.$$

616 Since $\Sigma_{XX} \succ 0$, the factor $\Sigma_{XX}^{-1/2}$ is invertible, so

$$\text{rank}(W_r^*) = \text{rank}\left(\Sigma_{YX} \Sigma_{XX}^{-1/2} V_r V_r^\top\right).$$

617 Let $\Sigma_{YX} \Sigma_{XX}^{-1/2} = U \text{diag}(\sigma_1, \dots, \sigma_{\text{rank}(\Sigma_{YX})}, 0, \dots, 0) V^\top$ be an SVD. By the identity

$$\Sigma_{XX}^{1/2} B_{\text{OLS}}^\top B_{\text{OLS}} \Sigma_{XX}^{1/2} = (\Sigma_{YX} \Sigma_{XX}^{-1/2})^\top (\Sigma_{YX} \Sigma_{XX}^{-1/2}),$$

618 the columns of V_r are the top- r right singular vectors of $\Sigma_{YX} \Sigma_{XX}^{-1/2}$. Since V is orthogonal, $V_r V_r^\top$
 619 projects onto the top- r right-singular subspace, so

$$\Sigma_{YX} \Sigma_{XX}^{-1/2} V_r V_r^\top = U \text{diag}(\sigma_1, \dots, \sigma_{\min(r, \text{rank}(\Sigma_{YX}))}, 0, \dots, 0) V^\top,$$

620 which has rank $\min\{r, \text{rank}(\Sigma_{YX})\}$. Using the fact that $\text{rank}(\Sigma_{YX} \Sigma_{XX}^{-1/2}) = \text{rank}(\Sigma_{YX})$ (again
 621 because $\Sigma_{XX}^{-1/2}$ is invertible) yields (5). $\square$

622 C.2 Proof of Rank Growth

623 *Proof of Proposition 3.4.* From (6),

$$G_{K+1} = G_K + \Sigma_{Y^{(K+1)}X}^\top \Sigma_{Y^{(K+1)}X}, \quad (24)$$

624 so for every $v \in \mathbb{R}^d$,

$$v^\top G_{K+1} v = v^\top G_K v + \|\Sigma_{Y^{(K+1)}X} v\|^2.$$

625 Both terms on the right are nonnegative, so $v^\top G_{K+1} v = 0$ holds if and only if both $v^\top G_K v = 0$
 626 and $\Sigma_{Y^{(K+1)}X} v = 0$. Hence

$$\ker(G_{K+1}) = \ker(G_K) \cap \ker(\Sigma_{Y^{(K+1)}X}) \subseteq \ker(G_K).$$

627 Therefore

$$\dim \ker(G_{K+1}) \leq \dim \ker(G_K).$$

628 Since $G_K \in \mathbb{R}^{d \times d}$, rank-nullity gives

$$\text{rank}(G_{K+1}) = d - \dim \ker(G_{K+1}) \geq d - \dim \ker(G_K) = \text{rank}(G_K).$$

629 Using $\text{rank}(G_K) = \text{rank}(\Sigma_{YX}) = r_*(K)$, we obtain $r_*(K+1) \geq r_*(K)$, proving (1).

630 For (2), suppose $\ker(G_K) \not\subseteq \ker(\Sigma_{Y^{(K+1)}X})$ and pick $v \in \ker(G_K) \setminus \ker(\Sigma_{Y^{(K+1)}X})$. Then
 631 $v \in \ker(G_K)$ but $v \notin \ker(G_{K+1})$, so the inclusion above is strict, giving $r_*(K+1) > r_*(K)$. $\square$

632 C.3 Proof of Heterogeneity and Identifiability

633 *Proof of Proposition 3.8.* On the assumed perturbation event, $\|\widehat{\Sigma}_{YX} - \Sigma_{YX}\|_{\text{op}} \leq \eta_n^{YX}(\delta)$. Since
 634 $\text{rank}(\Sigma_{YX}) = r_*(K)$, applying Lemma A.5 with $r = r_*(K)$, $\epsilon = \eta_n^{YX}(\delta)$, and Wedin gap
 635 $g_{r_*}(\Sigma_{YX}) = \sqrt{\lambda_{\text{het}}}$ (7) yields the claim. $\square$

636 C.4 Proof of Capacity Lower Bound

637 *Proof of Proposition 3.9.* For any pair (W, A) , define the end-to-end map

$$B := WA \in \mathbb{R}^{Kd \times D_x}.$$

638 Since $W \in \mathbb{R}^{Kd \times d}$ and $A \in \mathbb{R}^{d \times D_x}$, one has $\text{rank}(B) \leq d$. Conversely, every matrix $B \in \mathbb{R}^{Kd \times D_x}$
 639 with $\text{rank}(B) \leq d$ admits a factorization $B = WA$ for some such (W, A) . Hence

$$\inf_{W,A} \mathbb{E} [\|Y - WAZ\|_2^2] = \inf_{\text{rank}(B) \leq d} \mathbb{E} [\|Y - BZ\|_2^2]. \quad (25)$$

640 Next, by population least squares, $B_{\text{OLS}}^{\text{raw}} = \Sigma_{YZ}\Sigma_{ZZ}^{-1}$ is the unconstrained minimizer of $\mathbb{E}[\|Y -$
 641 $BZ\|_2^2]$. Therefore, for every $B \in \mathbb{R}^{Kd \times D_x}$,

$$\mathbb{E} [\|Y - BZ\|_2^2] = \mathbb{E} [\|Y - B_{\text{OLS}}^{\text{raw}}Z\|_2^2] + \left\| (B - B_{\text{OLS}}^{\text{raw}})\Sigma_{ZZ}^{1/2} \right\|_F^2.$$

642 Thus

$$\inf_{\text{rank}(B) \leq d} \mathbb{E} [\|Y - BZ\|_2^2] = \mathbb{E} [\|Y - B_{\text{OLS}}^{\text{raw}}Z\|_2^2] + \inf_{\text{rank}(B) \leq d} \left\| B\Sigma_{ZZ}^{1/2} - T_K \right\|_F^2.$$

643 Since $\Sigma_{ZZ}^{1/2}$ is invertible, the rank constraint remains $\text{rank}(B\Sigma_{ZZ}^{1/2}) \leq d$. By the Eckart-Young
 644 theorem [34] for Frobenius-norm low-rank approximation,

$$\inf_{\text{rank}(B) \leq d} \left\| B\Sigma_{ZZ}^{1/2} - T_K \right\|_F^2 = \sum_{j>d} \sigma_j(T_K)^2. \quad (26)$$

645 Combining this with (25) yields

$$\inf_{W,A} \mathbb{E} [\|Y - WAZ\|_2^2] = \mathbb{E} [\|Y - B_{\text{OLS}}^{\text{raw}}Z\|_2^2] + \sum_{j>d} \sigma_j(T_K)^2. \quad (27)$$

646 The claimed inequality follows immediately. $\square$

647 C.5 Proof of Post-Saturation Risk Decomposition

648 *Proof of Proposition 3.11.* By definition, $\mathcal{R}_K(\hat{V}_{K,d}) = \mathcal{R}_K^*(d) + (\mathcal{R}_K(\hat{V}_{K,d}) - \mathcal{R}_K^*(d))$. The first
 649 term equals the underbraced architectural floor by Proposition 3.9. For the second, combining
 650 Corollary A.7 with the subspace-Lipschitz Assumption 3.10 yields, with probability at least $1 - \delta$,

$$\mathcal{R}_K(\hat{V}_{K,d}) - \mathcal{R}_K^*(d) \leq L \|\sin \Theta(V_{K,d}, \hat{V}_{K,d})\|_{\text{op}} \leq \frac{2L\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K - K^*)}}. \quad (28)$$

651 Adding the two bounds yields (12). $\square$

652 C.6 Proof of Unified Risk Decomposition

653 *Proof of Theorem 3.15.* By the reduced-rank regression solution in Lemma A.2, using the whitening
 654 argument as in Proposition 3.9, $\mathcal{R}_K^*(r)$ is exactly the optimal population risk among rank- r end-to-end
 655 predictor. Next, on $E_T(K; \rho_n)$ (10), the Wedin perturbation bound gives

$$\|\sin \Theta(V_{K,r}, \hat{V}_{K,r})\|_{\text{op}} \leq \frac{\rho_n(\delta) \|T_K\|_{\text{op}}}{\Delta_{K,r}^{\text{abs}} - \rho_n(\delta) \|T_K\|_{\text{op}}} = \frac{\rho_n(\delta)}{\bar{\Delta}_{K,r} - \rho_n(\delta)},$$

656 because $\Delta_{K,r}^{\text{abs}} = \sigma_r(T_K) - \sigma_{r+1}(T_K)$, $\bar{\Delta}_{K,r} = \Delta_{K,r}^{\text{abs}} / \|T_K\|_{\text{op}}$, and $\rho_n(\delta) < \bar{\Delta}_{K,r}$. This is the
 657 same perturbation step used in Proposition A.6, now applied at rank r . By Assumption 3.10,

$$\mathcal{R}_K(\hat{V}_{K,r}) - \mathcal{R}_K^*(r) \leq L \|\sin \Theta(V_{K,r}, \hat{V}_{K,r})\|_{\text{op}}.$$

658 Substituting the Wedin bound into this inequality gives (13). $\square$

659 C.7 Proof of Predictive Isotropy

660 *Proof.* Let $T_K = U_K S_K V_K^\top$ be a compact singular value decomposition, where $U_K \in \mathbb{R}^{Kd \times d}$
 661 and $V_K \in \mathbb{R}^{D_x \times d}$ have orthonormal columns, and $S_K = \text{diag}(\sigma_1(T_K), \dots, \sigma_d(T_K))$. Since
 662 $\text{rank}(T_K) = d$, $P_K = T_K \tilde{Z}$ lies in $\text{col}(T_K) = \text{col}(U_K)$. Thus $U_K^\top P_K$ is the intrinsic coordinate
 663 vector of P_K on its rank- d predictive range. Covariance isotropy of P_K on this range is therefore
 664 equivalent to covariance isotropy of $U_K^\top P_K$ on $\mathbb{R}^d$. Using $P_K = T_K \tilde{Z}$ and $\text{Cov}(\tilde{Z}) = I_{D_x}$,

$$\text{Cov}(U_K^\top P_K) = \text{Cov}(U_K^\top T_K \tilde{Z}) = \text{Cov}(S_K V_K^\top \tilde{Z}) = S_K^2.$$

665 Thus P_K has isotropic covariance on its rank- d range if and only if $S_K^2 = cI_d$ for some $c > 0$. The
 666 nonzero eigenvalues of $H_K = T_K^\top T_K$ are the diagonal entries of S_K^2 , so this is exactly (14). $\square$

667 C.8 Proof of Predictive Isotropy Minimize the Excess-Risk Bound

668 *Proof of Theorem 4.2.* Let $\lambda_1 \geq \dots \geq \lambda_d > 0$ denote the nonzero eigenvalues of H_K , with
 669 $\sum_{i=1}^d \lambda_i = \tau$. At fixed $\text{tr}(H_K) = \tau$, $\lambda_d \leq \tau/d \leq \lambda_1$, with both equalities if and only if the spectrum
 670 is flat; hence the isotropic spectrum simultaneously minimizes λ_1 and maximizes $\lambda_d = \sigma_d(T_K)^2$,
 671 which uniquely maximizes $\kappa_K^2 = \lambda_d/\lambda_1 \leq 1$. At rank d , the **relative Wedin denominator** is
 672 $\bar{\Delta}_{K,d} = \sigma_d(T_K)/\|T_K\|_{\text{op}} = \kappa_K$. For a fixed **relative perturbation level** $\rho_n(\delta)$, the **excess-risk factor**
 673 $\rho_n(\delta)/(\kappa_K - \rho_n(\delta))$ is strictly decreasing in κ_K whenever $\rho_n(\delta) < \kappa_K$. Since the flat spectrum is
 674 the unique fixed-trace spectrum with $\kappa_K = 1$, it uniquely minimizes this bound. $\square$

675 C.9 Proof of Covariance Isotropy Subsumption

676 *Proof of Theorem 4.5.* By Lemma 4.4, $\Lambda_G(Z) = GV_K^\top \Sigma_{ZZ}^{-1/2} Z$, so

$$\text{Cov}(\Lambda_G(Z)) = GV_K^\top \Sigma_{ZZ}^{-1/2} \Sigma_{ZZ} \Sigma_{ZZ}^{-1/2} V_K G^\top = GV_K^\top V_K G^\top = GG^\top = I_d,$$

677 using $V_K^\top V_K = I_d$ from the SVD and $GG^\top = I_d$ by orthogonality property. $\square$

678 C.10 Proof of Linear Gaussian Gauge

679 *Proof of Corollary 4.7.* The whitened context $\tilde{Z} = \Sigma_{ZZ}^{-1/2} Z \sim \mathcal{N}(0, I_{D_x})$ under Gaussian context,
 680 and the rows of $V_K^\top$ are orthonormal, so $\Lambda_I Z = V_K^\top \tilde{Z} \sim \mathcal{N}(0, I_d)$. Hence $G^* = I$ realizes
 681 Theorem 4.6, and the encoder-predictor factorization specializes to (Λ_I, F_I) . $\square$

682 D Additional Experimental Details

683 D.1 Synthetic Experiment Details

684 **Synthetic generator.** Each synthetic instance is generated from a latent linear-Gaussian model. We
 685 sample a latent vector $s \in \mathbb{R}^{r_*}$ and observe

$$z = Cs + \sigma_z \epsilon_z, \quad y_k = a_k^\top s + \sigma_y \epsilon_k,$$

686 where $C \in \mathbb{R}^{D_x \times r_*}$ is a randomly sampled orthonormal context operator, a_k is the k th target direction,
 687 and the noise terms are independent isotropic Gaussians. The population predictive structure is known,
 688 but the context coordinates are not aligned with the canonical basis.

689 **Target stacks.** The target operators are constructed in latent space. In the saturation and bottleneck
 690 experiments, we use a complementary target stack whose directions progressively cover the r_* latent
 691 predictive coordinates. In the heterogeneity experiment, the construction parameter is chosen so that
 692 the induced population heterogeneity level λ_{het} is sampled approximately evenly. The plot uses the
 693 measured heterogeneity level directly on the horizontal axis. In the predictive-isotropy ablation, the
 694 target stack is constructed to control the spectrum directly. For a spectrum decay parameter $\rho \in (0, 1]$,
 695 we set

$$A^\top A = Q \text{diag}(\lambda_1, \dots, \lambda_d) Q^\top, \quad \lambda_i = \tau \frac{\rho^{i-1}}{\sum_{j=1}^d \rho^{j-1}},$$

696 where Q is a random orthogonal rotation and τ is fixed across all conditions. This keeps $\text{rank}(A) = d$
 697 and $\text{tr}(A^\top A) = \tau$ fixed, while the measured spectrum condition $\kappa_H^2 = \lambda_d(H_K)/\lambda_1(H_K)$ ranges
 698 from 1 to approximately 10^{-8} . Because the context operator has orthonormal columns and σ_z is
 699 fixed, the nonzero spectrum of $H_K = T_K^\top T_K$ changes only by the same context-noise scalar across
 700 all conditions. The decay parameter ρ is only a generator knob. The figure and interpretation use the
 701 measured operator quantities κ_H^2 and entropy gap.

702 **Models and estimators.** We first estimate the unrestricted linear predictor by ordinary least squares
 703 and then compute its rank- d reduced-rank regression approximation by truncating the singular
 704 spectrum. The learned BM-JEPA linear predictor is the best rank-constrained linear map induced by
 705 the finite training sample. Population curves use the population cross-covariance operator under the
 706 same target stack and bottleneck dimension. Figure 3 gives a direct reduction check for Proposition 3.1
 707 and Lemma 3.2: the population RRR coefficient is explicitly factorized as WA , its rank matches
 708 $\min\{r, \text{rank}(\Sigma_{YX})\}$, and the factorized predictor has the same loss as the closed-form RRR solution
 709 up to numerical precision.

710 **Metrics.** All paper-facing MSE panels report average per-target test mean-squared error, matching
 711 the normalization of $\mathcal{R}_K$ in (11). The CSV files also retain fixed-reference-stack diagnostics on
 712 a common reference target stack. These are auxiliary representation probes, not the theory-facing
 713 risk quantities. In Figure 5, per-target $\mathcal{R}_K$ evaluates the changing target stack used for training.
 714 Reference-stack MSE evaluates transfer of the recovered subspace to a common target stack. As
 715 heterogeneity increases, $\mathcal{R}_K$ can remain nearly constant or rise slightly because the task becomes less
 716 redundant; the reference-stack probe can decrease at the same time because the recovered subspace
 717 becomes more complete. The standard-error band of $\mathcal{R}_K$ also shrinks as target coverage spreads
 718 across more latent directions, so the per-target average is less dominated by a single shared direction.
 719 In Figure 8, panel (d) uses a related reference-stack probe: after estimating the top- d right singular

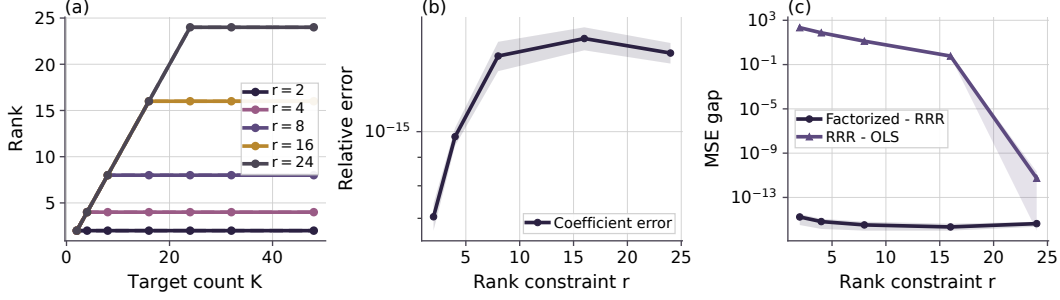


Figure 3: Reduction diagnostics. Panel (a) compares the rank of the closed-form RRR coefficient with $\min\{r, \text{rank}(\Sigma_{YX})\}$; dashed curves are the formula. Panels (b,c) show that an explicit WA factorization reproduces the RRR coefficient and loss up to roundoff.

subspace of the whitened operator $\hat{T}_K$, we fit a linear predictor from that recovered subspace to a fixed flat evaluation target stack. This reference-stack excess MSE is a representation diagnostic. It measures whether the recovered predictive subspace captures all latent directions equally well. Recovered rank is the effective numerical rank of the learned predictor, computed using a relative singular-value threshold of 0.05 times the leading singular value. The heterogeneity metric λ_{het} is the r_* th nonzero eigenvalue of the target-induced Gram operator. In the latent coordinates this is the same nonzero spectrum as $A^\top A$; in the observed context coordinates it is the same nonzero spectrum as $\Sigma_{ZY}\Sigma_{YZ}$ because $C^\top C = I$. Subspace error is measured by the largest principal-angle sine between the recovered and true predictive subspaces. For the heterogeneity recovery bound, we also compute the empirical perturbation $\hat{\epsilon} = \|\hat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\text{op}}$ and the corresponding Davis-Kahan/Wedin-style denominator $\sqrt{\lambda_{\text{het}}} - \hat{\epsilon}$. When this denominator is nonpositive, the finite-sample bound is vacuous. We keep these quantities in the CSV as validity diagnostics. We omit them from Figure 5 because the bound is loose for this sweep, and report the direct concentration check separately in Figure 9. The post-saturation experiment plots the recovery-bound diagnostic in a finite window where the relative-conditioning condition is non-vacuous.

For the bottleneck experiment, the plotted spectral-tail quantity is the reduced-rank approximation error predicted by the bottleneck spectral-tail result:

$$\mathcal{E}_{\text{tail}}(d) = \sum_{j>d} \sigma_j^2,$$

where $\{\sigma_j\}$ are singular values of the relevant whitened regression operator. In the figure we report the normalized ratio $\mathcal{E}_{\text{tail}}(d) / \sum_j \sigma_j^2$ and compare it with the correspondingly normalized observed excess risk of the rank- d estimator over the unrestricted OLS estimator.

The rank panel in the bottleneck figure reports the effective rank of the learned reduced-rank predictor, not the rank of the stacked cross-covariance. The latter is fixed by the target stack and does not change when the bottleneck dimension is swept. Since the synthetic target stack has latent predictive rank $r_* = 24$, a well-estimated rank- d predictor is expected to use approximately $\min\{d, r_*\}$ directions. The rank panel checks that the bottleneck cap is the active rank constraint before $d = r_*$ and that additional capacity is unused afterward.

Baselines. The main finite-sample baseline is unrestricted OLS, which removes the embedding bottleneck. Population curves are included as reference predictions for the same synthetic distribution. The bottleneck experiment additionally reports a plug-in spectral tail computed from the finite-sample OLS spectrum, so the risk decomposition can be checked using learned empirical quantities rather than only population quantities.

Configuration summary. We list the main Phase-1 configurations used for the reported figures in Table 2.

Additional theory diagnostics. The CSV schema also records the effective rank of the learned RRR coefficient, the finite-sample target value $\min\{d, \hat{r}_*(K)\}$ predicted by the rank identity, and

Table 2: Phase-1 synthetic experiment configurations.

Experiment	n_{train}	n_{test}	D_x	r_*	K	d	Noise
K saturation	1024	8192	160	64	sweep	64	0.12
Heterogeneity	32768	8192	64	24	48	24	0.10
Bottleneck	128	8192	64	24	48	sweep	0.16
Unified risk	sweep	8192	80	32	64	16	0.25
Post-saturation	16384	8192	160	64	sweep	64	0.25
Predictive isotropy	1024	8192	64	24	24	24	0.25
Gauge factorization	8192	8192	64	24	24	24	0.25

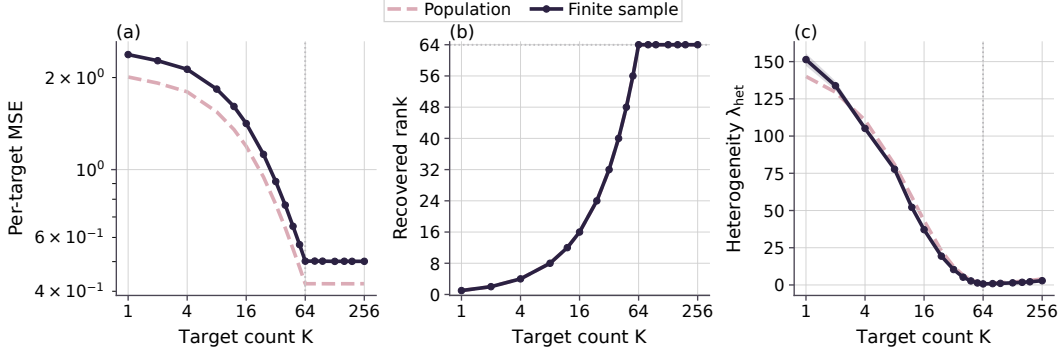


Figure 4: K -saturation diagnostic. Complementary targets reduce per-target MSE and increase recovered rank until predictive rank saturates.

755 their difference. This directly checks the rank identity beyond the displayed rank of the stacked
756 cross-covariance.

757 Figure 7 gives a diagnostic for Theorem 3.15 separate from the post-saturation special case. The
758 target stack has $K = 64$, $r_* = 32$, and a controlled rank- r spectral gap at $r = d = 16$, so the
759 spectral-tail term is nonzero while the retained gap is positive. The raw OLS floor is about 0.121, the
760 per-target spectral tail is about 0.244, and the population rank- r risk is therefore about 0.365. We
761 sweep $n_{\text{train}} \in \{512, 1024, 2048, 4096, 8192, 16384\}$ over ten seeds. The retained relative gap is
762 $\Delta_{K,r} = 0.408$. The calibrated $\hat{\rho}_{0.9}$ decreases from 0.291 at $n = 512$ to 0.049 at $n = 16384$, with
763 9/10 perturbation-event coverage at each sample size, so the gap-compatible relative-perturbation
764 condition is non-vacuous throughout this window. The empirical subspace-Lipschitz constant is
765 $\hat{L} = 7.29 \times 10^{-2}$, computed as the maximum of $[\hat{\mathcal{R}}_K - \mathcal{R}_K^*(r)]_+ / \|\sin \Theta_T(V_{K,r}, \hat{V}_{K,r})\|_{\text{op}}$ over
766 all plotted runs. The finite-sample per-target risk decreases from 0.392 to 0.366 and stays below the
767 calibrated theorem bound, while approaching the population rank- r floor.

768 The diagnostic in Figure 1 tests the post-saturation relative-conditioning mechanism. The target stack
769 is full rank from $K^* = d = 64$ onward, so additional targets do not change $r_*(K)$. The relevant
770 post-saturation conditioning ratio is

$$\kappa_K^2 = \frac{\lambda_d(H_K)}{\lambda_1(H_K)}, \quad H_K = T_K^\top T_K, \quad T_K = \Sigma_{YZ}^{(K)} \Sigma_{ZZ}^{-1/2}.$$

771 The initial saturated target stack is deliberately imbalanced. Writing $A_K^\top A_K$ for the latent target-stack
772 design Gram, we initialize

$$A_{K^*}^\top A_{K^*} = Q \text{diag}(c_1, \dots, c_{64}) Q^\top,$$

773 where Q is a fixed random orthogonal rotation and

$$c_i = 0.02^{(i-1)/63}.$$

774 The stack is already rank-64, but $\kappa_{K^*}^2 \approx 0.02$. For $K > K^*$, each additional target row is added to the
775 currently weakest eigendirection, with 0.1 units of target energy. This keeps $r_*(K) = d$ throughout

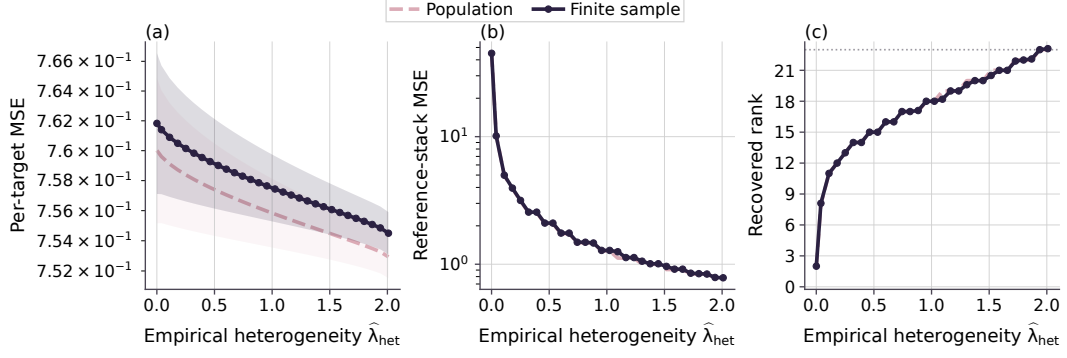


Figure 5: Target-heterogeneity diagnostic. Larger $\hat{\lambda}_{\text{het}}$ improves rank recovery; the reference-stack probe measures transfer to a common target stack.

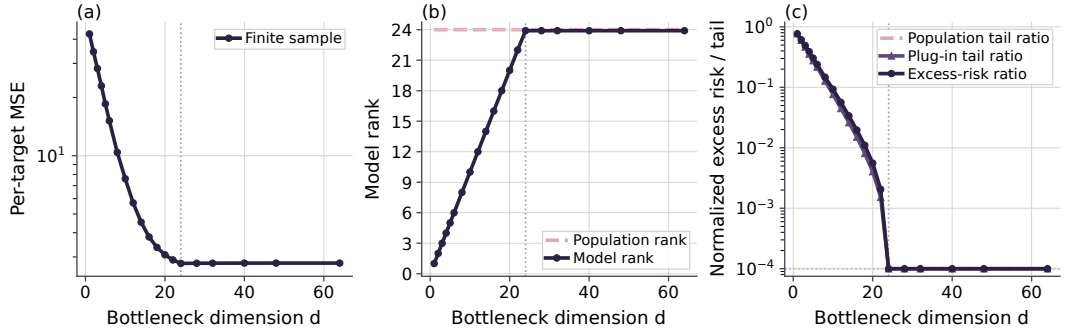


Figure 6: Embedding-bottleneck diagnostic. As d increases, model rank follows $\min\{d, r_\star\}$ and excess risk tracks the spectral-tail ratio.

776 the plotted window while increasing κ_K^2 . The finite-sample subspace error and per-target excess
777 MSE decrease over the same window, consistent with the post-saturation claim: extra targets help
778 only when they improve relative conditioning of the already-accessible predictive subspace. Figure 1
779 reports the weakest and top eigenvalues of $H_K = T_K^\top T_K$, the relative conditioning ratio κ_K^2 , the
780 effective rank of T_K , the risk-stability bound, excess risk, and finite-sample risk. The effective-rank
781 panel checks that $r(T_K) := \text{tr}(T_K^\top T_K) / \|T_K\|_{\text{op}}^2$ remains bounded over the finite window required
782 by the post-saturation relative-geometry condition. We compute the empirical risk-stability ratio

$$\hat{L}_K = \frac{[\hat{\mathcal{R}}_K - \mathcal{R}_K^*]_+}{\|\sin \Theta(V_K, \hat{V}_K)\|_{\text{op}}},$$

783 using per-target excess MSE in the numerator, and set $\hat{L} = \max_{K, \text{seed}} \hat{L}_K$. Panel (d) plots the excess-
784 risk left-hand side against $\hat{L} \|\sin \Theta(V_K, \hat{V}_K)\|_{\text{op}}$, matching the form of the normalized risk-stability
785 condition. Each seed is one deterministic draw of the synthetic data and target stack, and the same $\hat{L}$
786 is used across the full plotted window. In the current run this gives $\hat{L} = 7.07 \times 10^{-3}$.

787 For the relative-perturbation statement of Proposition 3.11, we set $\delta = 0.1$ and compute

$$\hat{\eta}_K = \frac{\|\hat{T}_K - T_K\|_{\text{op}}}{\|T_K\|_{\text{op}}}.$$

788 The plotted finite-window envelope is the empirical quantile $\hat{\rho}_{0.9}$ of $\hat{\eta}_K$ over the plotted (K, seed)
789 pairs. This is a calibrated relative-noise diagnostic. In the current run, $\hat{\rho}_{0.9} = 8.61 \times 10^{-2}$ and covers

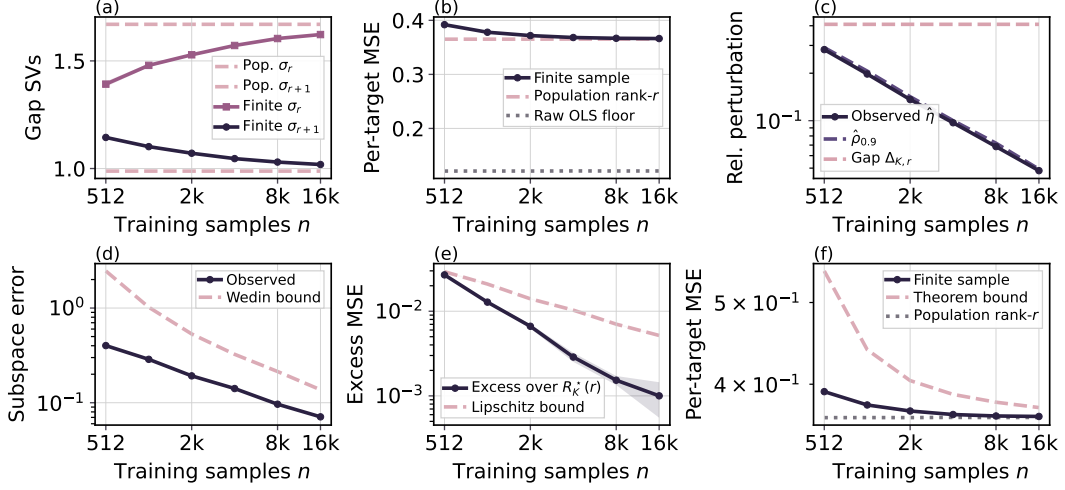


Figure 7: Unified-risk diagnostic for Theorem 3.15. The experiment fixes $r = 16 < r_* = 32$, so the population spectral tail is active. Panels check the retained gap, perturbation event, Wedin recovery, subspace-Lipschitz bridge, and final risk bound.

790 90/100 relative-perturbation samples by construction. Panel (e) uses the calibrated bound

$$\frac{2\widehat{L}\widehat{\rho}_{0.9}}{\kappa_K}$$

791 from the excess-risk term in (12). Panel (f) adds the same term to the population OLS risk estimate,
 792 matching the risk-decomposition bound in Proposition 3.11. Here $d = r_*(K)$, so the spectral-tail
 793 term is zero up to numerical precision.

794 We evaluate the empirical analogue of the half-gap condition stated in Proposition 3.11 using the
 795 normalized perturbation

$$\widehat{\eta}_K := \frac{\|\widehat{T}_K - T_K\|_{\text{op}}}{\|T_K\|_{\text{op}}}.$$

796 The condition $\widehat{\eta}_K \leq \kappa_K/2$ holds seed-wise for all deterministic runs in the plotted range. The
 797 weakest endpoint, $K = 64$, is deliberately closest to the small-conditioning boundary and remains
 798 just inside the half-gap window for the current sample size. Representative values are:

K	$\widehat{\eta}_{K,\text{mean}}$	$\widehat{\eta}_{K,\text{max}}$	$\kappa_K/2$	valid seeds
64	0.0613	0.0656	0.0707	10/10
96	0.0626	0.0688	0.1751	10/10
512	0.0929	0.0947	0.4748	10/10

799 **Concentration and perturbation diagnostics.** Figure 9 compares the empirical left-hand sides with
 800 calibrated right-hand sides for the concentration and relative-perturbation statements. Panel (a) targets
 801 the cross-covariance perturbation controlled by Lemma A.4. Let $U = (Y^\top, Z^\top)^\top$, $M = \mathbb{E}[UU^\top]$,
 802 and $r(M) = \text{tr}(M)/\|M\|_{\text{op}}$. The plotted inequality is

$$\|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\text{op}} \leq \widehat{C}_{0.9}\|M\|_{\text{op}} \left(\sqrt{\frac{r(M) + \log(1/\delta)}{n}} + \frac{r(M) + \log(1/\delta)}{n} \right),$$

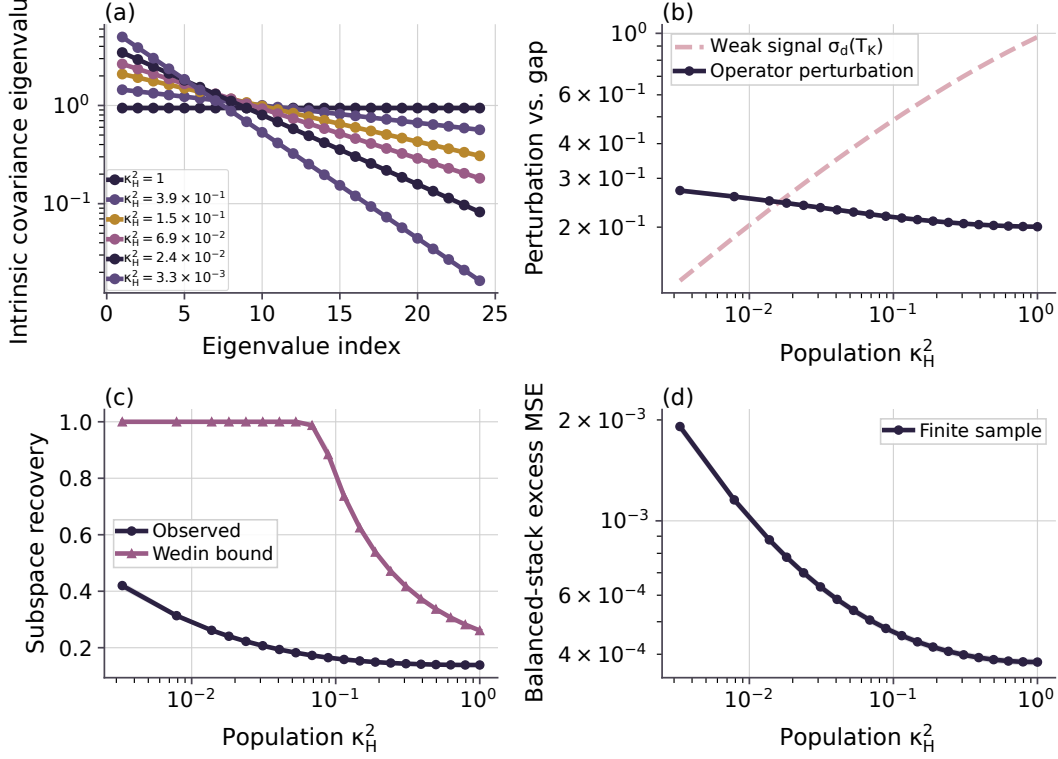


Figure 8: Predictive-isotropy diagnostic. At fixed rank and trace, flatter spectra of $H_K = T_K^\top T_K$ improve recovery and reduce balanced-stack excess MSE.

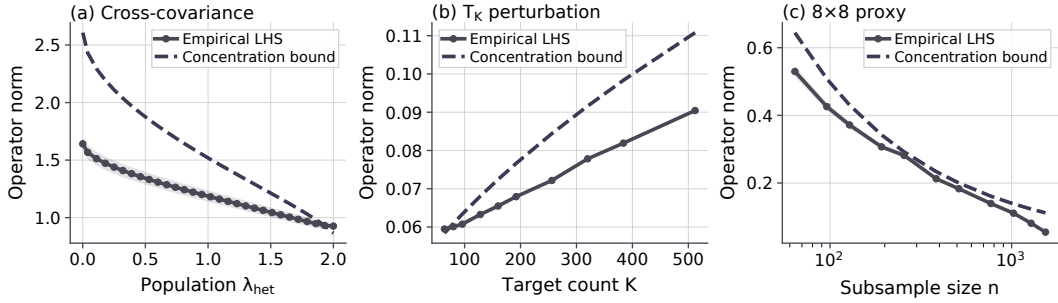


Figure 9: Concentration and perturbation diagnostics. Panels (a)–(c) compare the empirical cross-covariance perturbation, the T_K perturbation, and the real 8×8 digit T_K proxy with their calibrated bounds. Solid curves are empirical left-hand sides; dashed curves are calibrated right-hand sides.

803 evaluated on the heterogeneity sweep. Panel (b) gives the analogous finite-window perturbation
 804 diagnostic with T_K in place of Σ_{YZ} :

$$\|\hat{T}_K - T_K\|_{\text{op}} \leq \hat{C}_{0.9} \|T_K\|_{\text{op}} \left(\sqrt{\frac{r(T_K) + \log(1/\delta)}{n}} + \frac{r(T_K) + \log(1/\delta)}{n} \right).$$

805 Panel (c) repeats the T_K calculation on the real 8×8 digit halves. Since the real-data population
 806 operator is unknown, we use the full digit dataset to define a reference operator T_{ref} and compare
 807 subsample estimates $\hat{T}_n$ against it. We use $\delta = 0.1$ and set $\hat{C}_{0.9}$ by a 90% order-statistic calibration.

808 **Predictive-isotropy construction.** The fixed-trace predictive-isotropy experiment tests the second-
 809 moment prediction in Lemma 4.1 and the fixed-trace bound minimization in Theorem 4.2, with rank
 810 and total signal energy held fixed. All conditions use $D_x = 64$, $r_* = d = K = 24$, $\sigma_z = \sigma_y = 0.25$,

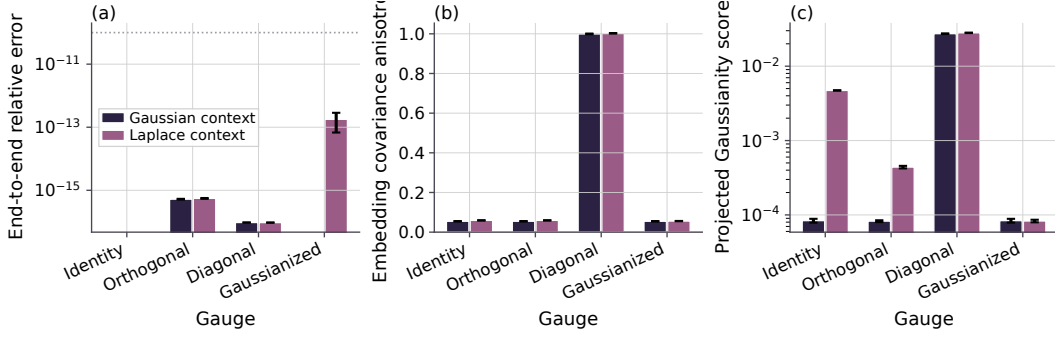


Figure 10: Gauge-factorization diagnostic. Gauge changes preserve prediction while changing encoder covariance and Gaussianity diagnostics.

811 and $\text{tr}(H_K) = 22.59$ after the common whitening/context-noise scaling. The generator uses 20
812 decay values between $\rho = 1.00$ and $\rho = 0.78$, concentrated near the transition where the weakest
813 retained signal becomes comparable to the empirical perturbation scale. These decay values induce a
814 dense sweep of population conditioning values $\kappa_H^2 \in [3.30 \times 10^{-3}, 1]$. Representative points are:

ρ	κ_H^2	entropy gap	$\ \sin \Theta_T\ _{\text{op}}$
1.00	1.0000	0.000	0.139
0.94	2.41×10^{-1}	-0.090	0.146
0.90	8.86×10^{-2}	-0.253	0.165
0.86	3.12×10^{-2}	-0.495	0.207
0.84	1.81×10^{-2}	-0.644	0.241
0.78	3.30×10^{-3}	-1.191	0.420

815 The trace column is omitted because it is constant by construction. Since $\tilde{Z} = \Sigma_{ZZ}^{-1/2} Z$ is Gaussian
816 in this generator, the intrinsic covariance of $P_K = T_K \tilde{Z}$ on its rank- d range has eigenvalues $\lambda_i(H_K)$.
817 Panel (a) of Figure 8 is the covariance spectrum appearing in Lemma 4.1. Panel (b) plots the two quan-
818 tities that determine the perturbative denominator: $\sigma_d(T_K)$ and $\|\hat{T}_K - T_K\|_{\text{op}}$. Panel (c) compares the
819 observed subspace recovery error with the Wedin bound $\|\hat{T}_K - T_K\|_{\text{op}} / (\sigma_d(T_K) - \|\hat{T}_K - T_K\|_{\text{op}})$,
820 clipped at one when the denominator is nonpositive. The small- κ_H^2 settings intentionally enter this
821 vacuous-bound regime. These settings mark the transition from perturbative recovery to the boundary
822 where weak directions are hard to recover. In the current ten-seed run, 16 of the 20 spectrum settings
823 have a positive perturbative denominator for every seed, and 10 of 20 satisfy the half-gap diagnostic
824 for every seed.

825 **Gauge-factorization diagnostic.** The gauge diagnostic tests Section 4.2. For each seed, we construct
826 the population whitened predictor $T_K = U_K S_K V_K^\top$ and compare identity, random orthogonal,
827 anisotropic diagonal, and nonlinear Gaussianizing gauges under both Gaussian and standardized
828 Laplace SVD-coordinate distributions. For an invertible linear gauge G , the synthetic encoder and
829 predictor are

$$\Lambda_G(Z) = G V_K^\top \Sigma_{ZZ}^{-1/2} Z, \quad F_G(\cdot) = U_K S_K G^{-1}(\cdot).$$

830 Panel (a) of Figure 10 checks Lemma 4.4: all gauges reproduce the same end-to-end prediction
831 up to numerical precision. Panel (b) checks the covariance part of Theorem 4.5: identity and
832 orthogonal gauges remain covariance-isotropic up to finite-sample error for both context laws, while
833 the diagonal gauge is deliberately anisotropic. Panel (c) reports a Cramer-Wold-style projected
834 standard-normal discrepancy. For Gaussian context, the identity/SVD-aligned gauge already realizes
835 the linear-Gaussian special case in Corollary 4.7. For Laplace context, identity remains covariance-
836 isotropic but not Gaussian; the nonlinear Gaussianizing gauge maps the intrinsic coordinates to
837 standard-normal marginals and sharply reduces the projected discrepancy. This matches the nonlinear
838 gauge construction in Theorem 4.6.

839 D.2 Real-Image Regularizer Details

840 **Regularization implication.** The theory motivates a prediction-aware analogue of embedding Gaus-
 841 sian regularization. Standard embedding regularizers act on the marginal pushforward $\mathcal{L}(AZ)$. The
 842 theory-facing object here is the intrinsic end-to-end predictive pushforward from Lemma 4.1. For a
 843 minibatch, project the learned predicted outputs onto the learned rank- d predictive range and collect
 844 the intrinsic coordinates into rows $\hat{p}_n^{\text{int}}$ of

$$\hat{P}^{\text{int}} \in \mathbb{R}^{N \times d}.$$

845 The proposed regularizer samples random directions $\mathcal{A} \subset \mathbb{S}^{d-1}$ and averages a univariate Gaussianity
 846 test, such as the Epps-Pulley statistic, over the projected samples:

$$\mathcal{R}_{\text{PI}}(\hat{P}^{\text{int}}; \mathcal{A}) = \frac{1}{|\mathcal{A}|} \sum_{u \in \mathcal{A}} \mathcal{T}(\{u^\top \hat{p}_n^{\text{int}}\}_{n=1}^N).$$

847 This is a Cramer-Wold-style, prediction-aware counterpart to SIGReg: it acts on intrinsic predictor
 848 outputs rather than encoder embeddings. Matching projected standard normals is stronger than the
 849 second-moment condition $H_K = \alpha I$, but it implies the isotropic predictive covariance targeted by
 850 Lemma 4.1. The nonlinear diagnostic uses the objective

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{emb}} \mathcal{L}_{\text{emb-iso}} + \lambda_{\text{pred}} \mathcal{R}_{\text{PI}}.$$

851 The embedding term prevents marginal representation collapse. The prediction term targets the
 852 predictive spectrum appearing in the finite-sample recovery and excess-risk bounds. The gauge
 853 results explain why a sufficiently strong prediction-aware regularizer can subsume Encoder isotropy
 854 at the population level. The combined objective tests whether the Encoder term adds useful finite-
 855 sample optimization or collapse-prevention effects.

856 **Real-image regularizer experiments.** We run two learned nonlinear left-to-right digit-completion
 857 diagnostics. In both cases, the context is the left half of the image and the target is the right half. The
 858 small-data `sklearn` task uses 8×8 handwritten digits, an MLP encoder, a 16-dimensional predictor
 859 representation, and a linear pixel head. The scaled MNIST task uses 28×28 images, a convolutional
 860 encoder, a 128-dimensional predictor representation, and a spatial convolutional decoder. We identify
 861 the predictor representation with the finite-network analogue of $\hat{P}_K^{\text{int}}$ in (16); the pixel head or decoder
 862 is the readout used for target prediction.

863 We compare four objectives: no regularizer, Encoder Gaussianity, prediction-side Gaussianity, and
 864 prediction-plus-encoder Gaussianity. The projected-Gaussianity term combines a Cramer-Wold/Epps-
 865 Pulley statistic with a covariance-to-identity penalty and is applied either to the encoder embedding or
 866 to the predictor representation. The combined condition is a control: if the prediction-side condition
 867 captures the risk-relevant pushforward, adding the Encoder auxiliary term should not further improve
 868 prediction.

869 MSE is the mean squared error over right-half target pixels after normalizing image intensities to
 870 $[0, 1]$. RMSE is measured on the same gray-scale range. For MNIST, we also report foreground MSE
 871 over right-half pixels with intensity larger than 0.05, because background pixels dominate the raw
 872 target-half average. The Gaussianity score is a diagnostic distance to projected standard-normal laws,
 873 and effective rank is a conditioning diagnostic; the primary outcome is still target prediction risk.

874 Figures 11 and 12 and Table 1 report the real-image diagnostics. On the low-label 8×8 task,
 875 prediction-side regularization reduces test MSE from 9.23×10^{-2} to 6.00×10^{-2} and train-test gap
 876 from 8.54×10^{-2} to 2.00×10^{-2} . It reduces predictive Gaussianity distance from 1.14×10^{-1} to
 877 4.33×10^{-2} and raises predictive effective rank from 4.13 to 6.41. The prediction-only and prediction-
 878 plus-encoder conditions have nearly identical test MSE; the combined condition is slower.

879 The MNIST CNN diagnostic uses 10000 training examples, the standard 10000-example MNIST
 880 test split, 50 epochs, batch size 256, learning rate 3×10^{-4} with cosine decay, weight decay 10^{-4} ,
 881 convolutional channels (64, 128, 256), embedding dimension 256, predictor dimension 256, regular-
 882 izer weight 0.08, covariance weight 1.0, mixed precision when CUDA is available, and 128 random
 883 projection directions. In the ten-seed GPU run, prediction-side regularization improves test MSE

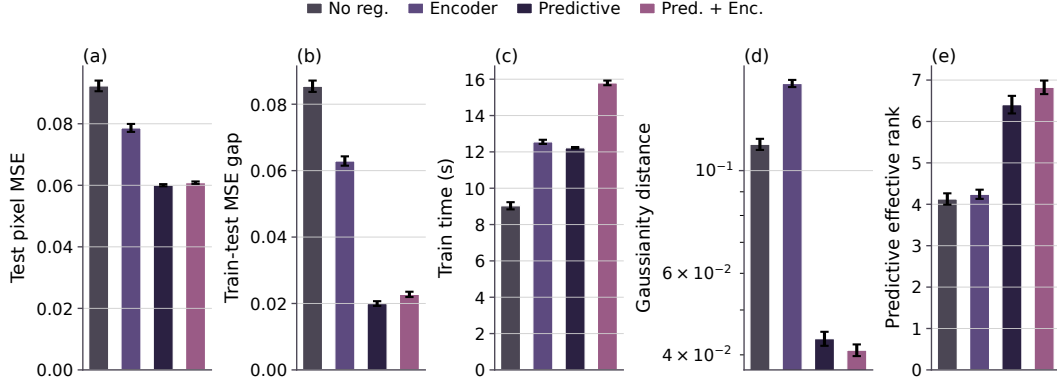


Figure 11: 8×8 digit regularizer diagnostic. Prediction-side regularization improves target-half MSE and prediction-side conditioning.

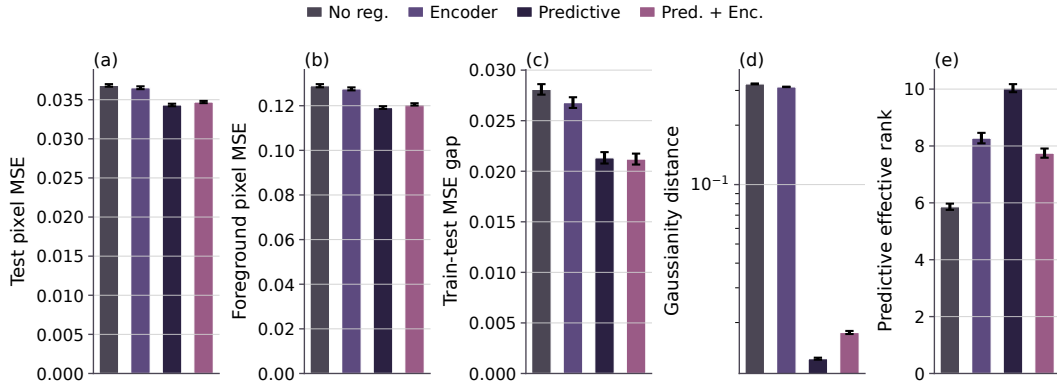


Figure 12: MNIST CNN regularizer diagnostic. The diagnostic reports target-half risk, foreground risk, prediction-side Gaussianity, and training time.

884 from 3.68×10^{-2} to 3.43×10^{-2} and foreground MSE from 1.29×10^{-1} to 1.19×10^{-1} . It reduces
 885 prediction-side Gaussianity distance from 3.24×10^{-1} to 1.31×10^{-2} and raises predictive effective
 886 rank from 5.86 to 10.03. The combined condition gives similar risk and the smallest train-test gap.
 887 Figures 2, 13 and 14 show completion examples: the left half of each held-out digit is observed, and
 888 each method predicts the right half.

889 **Reproducibility.** Synthetic experiments, the 8×8 digit regularizer diagnostic, and the MNIST CNN
 890 regularizer diagnostic are run with ten deterministic seeds. The synthetic suite and 8×8 diagnostic
 891 can be reproduced with

```
python -m experiments.run_all
```

892 followed by

```
python -m plots.plot_main.
```

893 The qualitative digit reconstruction figure is generated separately with

```
python -m plots.plot_regularizer_digits_samples.
```

894 The MNIST diagnostic can also be run directly with

```
python -m experiments.exp_regularizer_mnist_gpu.
```

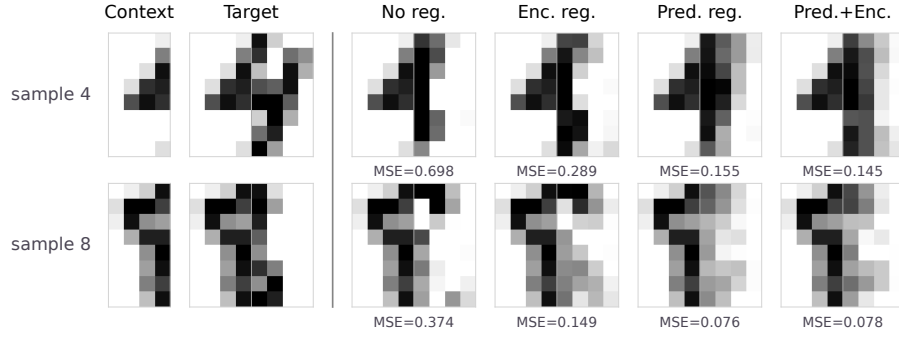


Figure 13: 8×8 digit completion examples. The left half is observed, and each method predicts the right half.

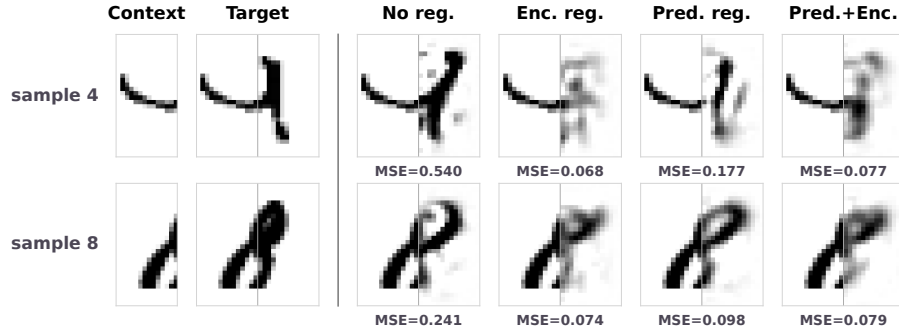


Figure 14: MNIST MLP completion examples. The left half is observed, and each method predicts the right half.

895 The scripts write per-experiment CSV files, the combined result table, and the main and appendix PDF
 896 figures. Unit tests cover deterministic generation, OLS/RRR behavior, rank constraints, subspace
 897 metrics, CSV schema, and figure-reference existence.

898 **Visualization choices.** Finite-sample estimates are plotted as solid curves with standard-error bands.
 899 Population predictions are plotted as dashed curves with standard-error bands when the population
 900 quantity varies across deterministic seeds. Bar charts show standard-error caps and seed-level points.
 901 For the bottleneck spectral-tail-ratio panel, values below 10^{-4} are visually floored in the log-scale
 902 display to keep the super-threshold regime legible; this affects only the display.

903 **NeurIPS Paper Checklist**

904 **1. Claims**

905 Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s
906 contributions and scope?

907 Answer: [\[Yes\]](#)

908 Justification: The main claims are scoped to the linearized/frozen-teacher setting and matched to
909 the stated theorems.

910 Guidelines:

- 911 • The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in
912 the paper.
- 913 • The abstract and/or introduction should clearly state the claims made, including the contribu-
914 tions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to
915 this question will not be perceived well by the reviewers.
- 916 • The claims made should match theoretical and experimental results, and reflect how much the
917 results can be expected to generalize to other settings.
- 918 • It is fine to include aspirational goals as motivation as long as it is clear that these goals are not
919 attained by the paper.

920 **2. Limitations**

921 Question: Does the paper discuss the limitations of the work performed by the authors?

922 Answer: [\[Yes\]](#)

923 Justification: Please see the Limitations paragraph at the end of Appendix B (Related Works and
924 Discussion).

925 Guidelines: Please see our limitation discussion in appendix.

- 926 • The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the
927 paper has limitations, but those are not discussed in the paper.
- 928 • The authors are encouraged to create a separate “Limitations” section in their paper.
- 929 • The paper should point out any strong assumptions and how robust the results are to violations of
930 these assumptions (e.g., independence assumptions, noiseless settings, model well-specification,
931 asymptotic approximations only holding locally). The authors should reflect on how these
932 assumptions might be violated in practice and what the implications would be.
- 933 • The authors should reflect on the scope of the claims made, e.g., if the approach was only tested
934 on a few datasets or with a few runs. In general, empirical results often depend on implicit
935 assumptions, which should be articulated.
- 936 • The authors should reflect on the factors that influence the performance of the approach. For
937 example, a facial recognition algorithm may perform poorly when image resolution is low or
938 images are taken in low lighting. Or a speech-to-text system might not be used reliably to
939 provide closed captions for online lectures because it fails to handle technical jargon.
- 940 • The authors should discuss the computational efficiency of the proposed algorithms and how
941 they scale with dataset size.
- 942 • If applicable, the authors should discuss possible limitations of their approach to address
943 problems of privacy and fairness.
- 944 • While the authors might fear that complete honesty about limitations might be used by reviewers
945 as grounds for rejection, a worse outcome might be that reviewers discover limitations that
946 aren’t acknowledged in the paper. The authors should use their best judgment and recognize
947 that individual actions in favor of transparency play an important role in developing norms
948 that preserve the integrity of the community. Reviewers will be specifically instructed to not
949 penalize honesty concerning limitations.

950 **3. Theory assumptions and proofs**

951 Question: For each theoretical result, does the paper provide the full set of assumptions and a
952 complete (and correct) proof?

953 Answer: [\[Yes\]](#)

954 Justification: Please see our main text and appendix.

955 Guidelines:

- 956 • The answer [N/A] means that the paper does not include theoretical results.
- 957 • All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- 958 • All assumptions should be clearly stated or referenced in the statement of any theorems.
- 959 • The proofs can either appear in the main paper or the supplemental material, but if they appear
- 960 in the supplemental material, the authors are encouraged to provide a short proof sketch to
- 961 provide intuition.
- 962 • Inversely, any informal proof provided in the core of the paper should be complemented by
- 963 formal proofs provided in appendix or supplemental material.
- 964 • Theorems and Lemmas that the proof relies upon should be properly referenced.

965 **4. Experimental result reproducibility**

966 Question: Does the paper fully disclose all the information needed to reproduce the main experi-

967 mental results of the paper to the extent that it affects the main claims and/or conclusions of the

968 paper (regardless of whether the code and data are provided or not)?

969 Answer: [Yes]

970 Justification: Please see our experimental section.

971 Guidelines:

- 972 • The answer [N/A] means that the paper does not include experiments.
- 973 • If the paper includes experiments, a [No] answer to this question will not be perceived well by
- 974 the reviewers: Making the paper reproducible is important, regardless of whether the code and
- 975 data are provided or not.
- 976 • If the contribution is a dataset and/or model, the authors should describe the steps taken to
- 977 make their results reproducible or verifiable.
- 978 • Depending on the contribution, reproducibility can be accomplished in various ways. For
- 979 example, if the contribution is a novel architecture, describing the architecture fully might
- 980 suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary
- 981 to either make it possible for others to replicate the model with the same dataset, or provide
- 982 access to the model. In general, releasing code and data is often one good way to accomplish
- 983 this, but reproducibility can also be provided via detailed instructions for how to replicate the
- 984 results, access to a hosted model (e.g., in the case of a large language model), releasing of a
- 985 model checkpoint, or other means that are appropriate to the research performed.
- 986 • While NeurIPS does not require releasing code, the conference does require all submissions to
- 987 provide some reasonable avenue for reproducibility, which may depend on the nature of the
- 988 contribution. For example
 - 989 (a) If the contribution is primarily a new algorithm, the paper should make it clear how to
 - 990 reproduce that algorithm.
 - 991 (b) If the contribution is primarily a new model architecture, the paper should describe the
 - 992 architecture clearly and fully.
 - 993 (c) If the contribution is a new model (e.g., a large language model), then there should either
 - 994 be a way to access this model for reproducing the results or a way to reproduce the model
 - 995 (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - 996 (d) We recognize that reproducibility may be tricky in some cases, in which case authors are
 - 997 welcome to describe the particular way they provide for reproducibility. In the case of
 - 998 closed-source models, it may be that access to the model is limited in some way (e.g.,
 - 999 to registered users), but it should be possible for other researchers to have some path to
 - 1000 reproducing or verifying the results.

1001 **5. Open access to data and code**

1002 Question: Does the paper provide open access to the data and code, with sufficient instructions to

1003 faithfully reproduce the main experimental results, as described in supplemental material?

1004 Answer: [Yes]

1005 Justification: We provide an anonymized supplementary code archive containing experiment
1006 scripts, configurations, result CSVs, plotting scripts, tests, and reproduction instructions. The
1007 experiments use public datasets (sklearn digits and MNIST) and synthetic generators included
1008 in the archive.

1009 Guidelines:

- 1010 • The answer [N/A] means that paper does not include experiments requiring code.
- 1011 • Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- 1012 • While we encourage the release of code and data, we understand that this might not be possible,
1013 so [No] is an acceptable answer. Papers cannot be rejected simply for not including code,
1014 unless this is central to the contribution (e.g., for a new open-source benchmark).
- 1015 • The instructions should contain the exact command and environment needed to run to reproduce
1016 the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- 1017 • The authors should provide instructions on data access and preparation, including how to access
1018 the raw data, preprocessed data, intermediate data, and generated data, etc.
- 1019 • The authors should provide scripts to reproduce all experimental results for the new proposed
1020 method and baselines. If only a subset of experiments are reproducible, they should state which
1021 ones are omitted from the script and why.
- 1022 • At submission time, to preserve anonymity, the authors should release anonymized versions (if
1023 applicable).
- 1024 • Providing as much information as possible in supplemental material (appended to the paper) is
1025 recommended, but including URLs to data and code is permitted.

1028 6. Experimental setting/details

1029 Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters,
1030 how they were chosen, type of optimizer) necessary to understand the results?

1031 Answer: [Yes]

1032 Justification: Please see our experimental section.

1033 Guidelines:

- 1034 • The answer [N/A] means that the paper does not include experiments.
- 1035 • The experimental setting should be presented in the core of the paper to a level of detail that is
1036 necessary to appreciate the results and make sense of them.
- 1037 • The full details can be provided either with the code, in appendix, or as supplemental material.

1038 7. Experiment statistical significance

1039 Question: Does the paper report error bars suitably and correctly defined or other appropriate
1040 information about the statistical significance of the experiments?

1041 Answer: [Yes]

1042 Justification: Please see our experimental section.

1043 Guidelines:

- 1044 • The answer [N/A] means that the paper does not include experiments.
- 1045 • The authors should answer [Yes] if the results are accompanied by error bars, confidence
1046 intervals, or statistical significance tests, at least for the experiments that support the main
1047 claims of the paper.
- 1048 • The factors of variability that the error bars are capturing should be clearly stated (for example,
1049 train/test split, initialization, random drawing of some parameter, or overall run with given
1050 experimental conditions).
- 1051 • The method for calculating the error bars should be explained (closed form formula, call to a
1052 library function, bootstrap, etc.)
- 1053 • The assumptions made should be given (e.g., Normally distributed errors).
- 1054 • It should be clear whether the error bar is the standard deviation or the standard error of the
1055 mean.

1056 • It is OK to report 1-sigma error bars, but one should state it. The authors should preferably
 1057 report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of
 1058 errors is not verified.

1059 • For asymmetric distributions, the authors should be careful not to show in tables or figures
 1060 symmetric error bars that would yield results that are out of range (e.g., negative error rates).

1061 • If error bars are reported in tables or plots, the authors should explain in the text how they were
 1062 calculated and reference the corresponding figures or tables in the text.

1063 **8. Experiments compute resources**

1064 Question: For each experiment, does the paper provide sufficient information on the computer
 1065 resources (type of compute workers, memory, time of execution) needed to reproduce the experi-
 1066 ments?

1067 Answer: [Yes]

1068 Justification: Please see our experimental section.

1069 Guidelines:

1070 • The answer [N/A] means that the paper does not include experiments.

1071 • The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud
 1072 provider, including relevant memory and storage.

1073 • The paper should provide the amount of compute required for each of the individual experi-
 1074 mental runs as well as estimate the total compute.

1075 • The paper should disclose whether the full research project required more compute than the
 1076 experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it
 1077 into the paper).

1078 **9. Code of ethics**

1079 Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS
 1080 Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

1081 Answer: [Yes]

1082 Justification: All authors read through carefully.

1083 Guidelines:

1084 • The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

1085 • If the authors answer [No], they should explain the special circumstances that require a deviation
 1086 from the Code of Ethics.

1087 • The authors should make sure to preserve anonymity (e.g., if there is a special consideration
 1088 due to laws or regulations in their jurisdiction).

1089 **10. Broader impacts**

1090 Question: Does the paper discuss both potential positive societal impacts and negative societal
 1091 impacts of the work performed?

1092 Answer: [Yes]

1093 Justification: Please see our impact statement.

1094 Guidelines:

1095 • The answer [N/A] means that there is no societal impact of the work performed.

1096 • If the authors answer [N/A] or [No], they should explain why their work has no societal impact
 1097 or why the paper does not address societal impact.

1098 • Examples of negative societal impacts include potential malicious or unintended uses (e.g.,
 1099 disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deploy-
 1100 ment of technologies that could make decisions that unfairly impact specific groups), privacy
 1101 considerations, and security considerations.

1102 • The conference expects that many papers will be foundational research and not tied to par-
 1103 ticular applications, let alone deployments. However, if there is a direct path to any negative
 1104 applications, the authors should point it out. For example, it is legitimate to point out that
 1105 an improvement in the quality of generative models could be used to generate Deepfakes for
 1106 disinformation. On the other hand, it is not needed to point out that a generic algorithm for
 1107 optimizing neural networks could enable people to train models that generate Deepfakes faster.

- 1108 • The authors should consider possible harms that could arise when the technology is being used
1109 as intended and functioning correctly, harms that could arise when the technology is being used
1110 as intended but gives incorrect results, and harms following from (intentional or unintentional)
1111 misuse of the technology.
- 1112 • If there are negative societal impacts, the authors could also discuss possible mitigation
1113 strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms
1114 for monitoring misuse, mechanisms to monitor how a system learns from feedback over time,
1115 improving the efficiency and accessibility of ML).

1116 11. Safeguards

1117 Question: Does the paper describe safeguards that have been put in place for responsible release
1118 of data or models that have a high risk for misuse (e.g., pre-trained language models, image
1119 generators, or scraped datasets)?

1120 Answer: [N/A]

1121 Justification: The paper releases no pretrained models, datasets, or other assets that could pose
1122 misuse or dual-use risks.

1123 Guidelines:

- 1124 • The answer [N/A] means that the paper poses no such risks.
- 1125 • Released models that have a high risk for misuse or dual-use should be released with necessary
1126 safeguards to allow for controlled use of the model, for example by requiring that users adhere
1127 to usage guidelines or restrictions to access the model or implementing safety filters.
- 1128 • Datasets that have been scraped from the Internet could pose safety risks. The authors should
1129 describe how they avoided releasing unsafe images.
- 1130 • We recognize that providing effective safeguards is challenging, and many papers do not require
1131 this, but we encourage authors to take this into account and make a best faith effort.

1132 12. Licenses for existing assets

1133 Question: Are the creators or original owners of assets (e.g., code, data, models), used in the
1134 paper, properly credited and are the license and terms of use explicitly mentioned and properly
1135 respected?

1136 Answer: [Yes]

1137 Justification: Please see our experimental section.

1138 Guidelines:

- 1139 • The answer [N/A] means that the paper does not use existing assets.
- 1140 • The authors should cite the original paper that produced the code package or dataset.
- 1141 • The authors should state which version of the asset is used and, if possible, include a URL.
- 1142 • The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- 1143 • For scraped data from a particular source (e.g., website), the copyright and terms of service of
1144 that source should be provided.
- 1145 • If assets are released, the license, copyright information, and terms of use in the package should
1146 be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for
1147 some datasets. Their licensing guide can help determine the license of a dataset.
- 1148 • For existing datasets that are re-packaged, both the original license and the license of the
1149 derived asset (if it has changed) should be provided.
- 1150 • If this information is not available online, the authors are encouraged to reach out to the asset's
1151 creators.

1152 13. New assets

1153 Question: Are new assets introduced in the paper well documented and is the documentation
1154 provided alongside the assets?

1155 Answer: [Yes]

1156 Justification: The supplementary code archive is documented with a README covering de-
1157 pendencies, reproduction commands, generated outputs, and dataset usage. The paper does not
1158 introduce a new dataset or pretrained model.

1159 Guidelines:

1160 • The answer [N/A] means that the paper does not release new assets.

1161 • Researchers should communicate the details of the dataset/code/model as part of their sub-

1162 missions via structured templates. This includes details about training, license, limitations,

1163 etc.

1164 • The paper should discuss whether and how consent was obtained from people whose asset is

1165 used.

1166 • At submission time, remember to anonymize your assets (if applicable). You can either create

1167 an anonymized URL or include an anonymized zip file.

1168 **14. Crowdsourcing and research with human subjects**

1169 Question: For crowdsourcing experiments and research with human subjects, does the paper

1170 include the full text of instructions given to participants and screenshots, if applicable, as well as

1171 details about compensation (if any)?

1172 Answer: [N/A]

1173 Justification: The paper does not involve crowdsourcing or research with human subjects.

1174 Guidelines:

1175 • The answer [N/A] means that the paper does not involve crowdsourcing nor research with

1176 human subjects.

1177 • Including this information in the supplemental material is fine, but if the main contribution of

1178 the paper involves human subjects, then as much detail as possible should be included in the

1179 main paper.

1180 • According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or

1181 other labor should be paid at least the minimum wage in the country of the data collector.

1182 **15. Institutional review board (IRB) approvals or equivalent for research with human subjects**

1183 Question: Does the paper describe potential risks incurred by study participants, whether such

1184 risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals

1185 (or an equivalent approval/review based on the requirements of your country or institution) were

1186 obtained?

1187 Answer: [N/A]

1188 Justification: The paper does not involve research with human subjects, so no IRB approval was

1189 required.

1190 Guidelines:

1191 • The answer [N/A] means that the paper does not involve crowdsourcing nor research with

1192 human subjects.

1193 • Depending on the country in which research is conducted, IRB approval (or equivalent) may be

1194 required for any human subjects research. If you obtained IRB approval, you should clearly

1195 state this in the paper.

1196 • We recognize that the procedures for this may vary significantly between institutions and

1197 locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines

1198 for their institution.

1199 • For initial submissions, do not include any information that would break anonymity (if applica-

1200 ble), such as the institution conducting the review.

1201 **16. Declaration of LLM usage**

1202 Question: Does the paper describe the usage of LLMs if it is an important, original, or non-

1203 standard component of the core methods in this research? Note that if the LLM is used only for

1204 writing, editing, or formatting purposes and does *not* impact the core methodology, scientific

1205 rigor, or originality of the research, declaration is not required.

1206 Answer: [N/A]

1207 Justification: The core contributions of this paper are theoretical (a reduced-rank regression formu-

1208 lation of JEPA, the predictive isotropy condition, and accompanying proofs). LLM-based coding

1209 assistants were used only as basic code assistance for implementing the numerical validation

1210 experiments (e.g., plotting and data pipelines), and did not impact the core methodology, scientific

1211 rigor, or originality of the research. Per NeurIPS policy, declaration is therefore not required.

- 1212 Guidelines:
- 1213 • The answer [N/A] means that the core method development in this research does not involve
- 1214 LLMs as any important, original, or non-standard components.
- 1215 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not be
- 1216 described.